

BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



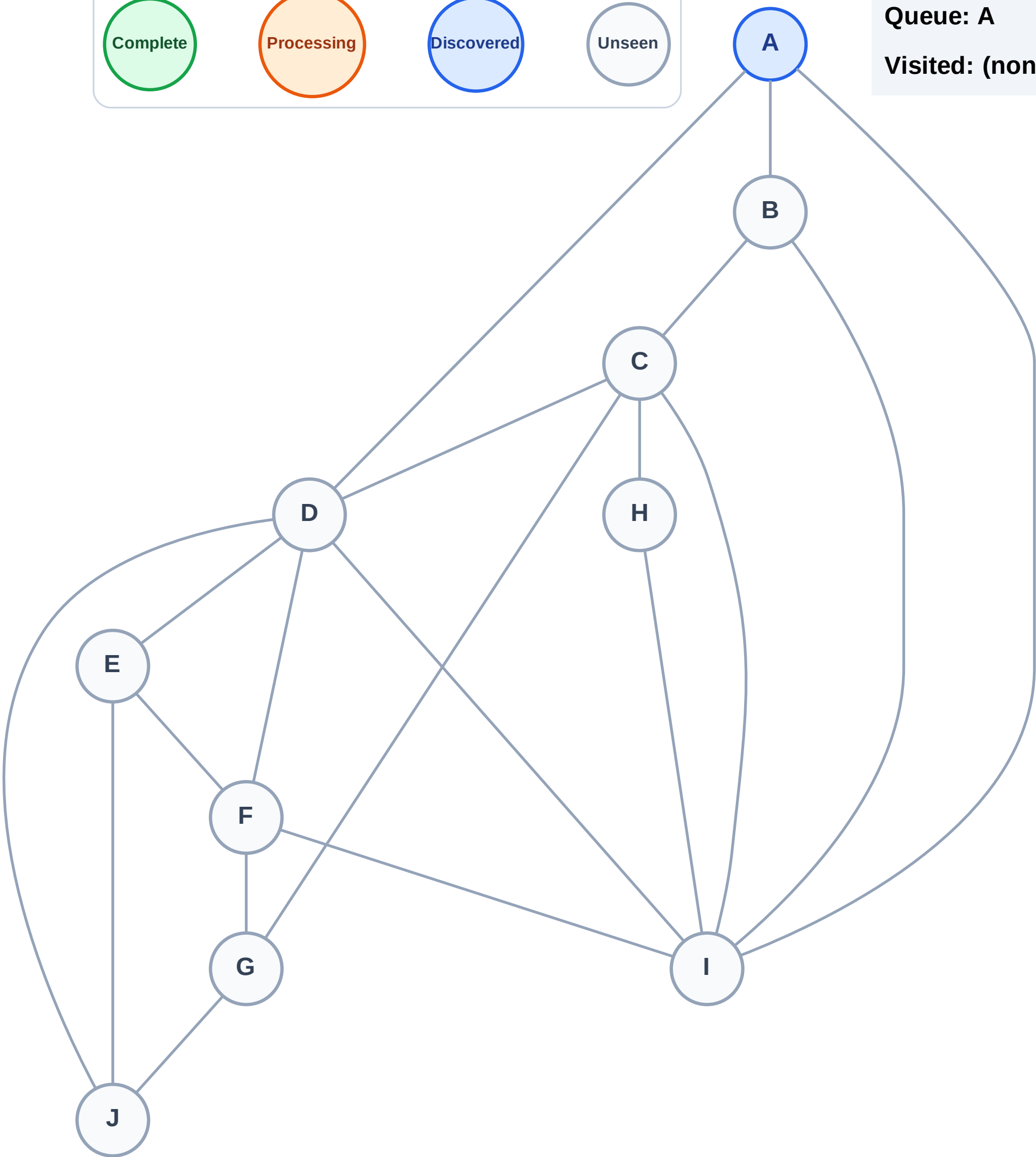
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

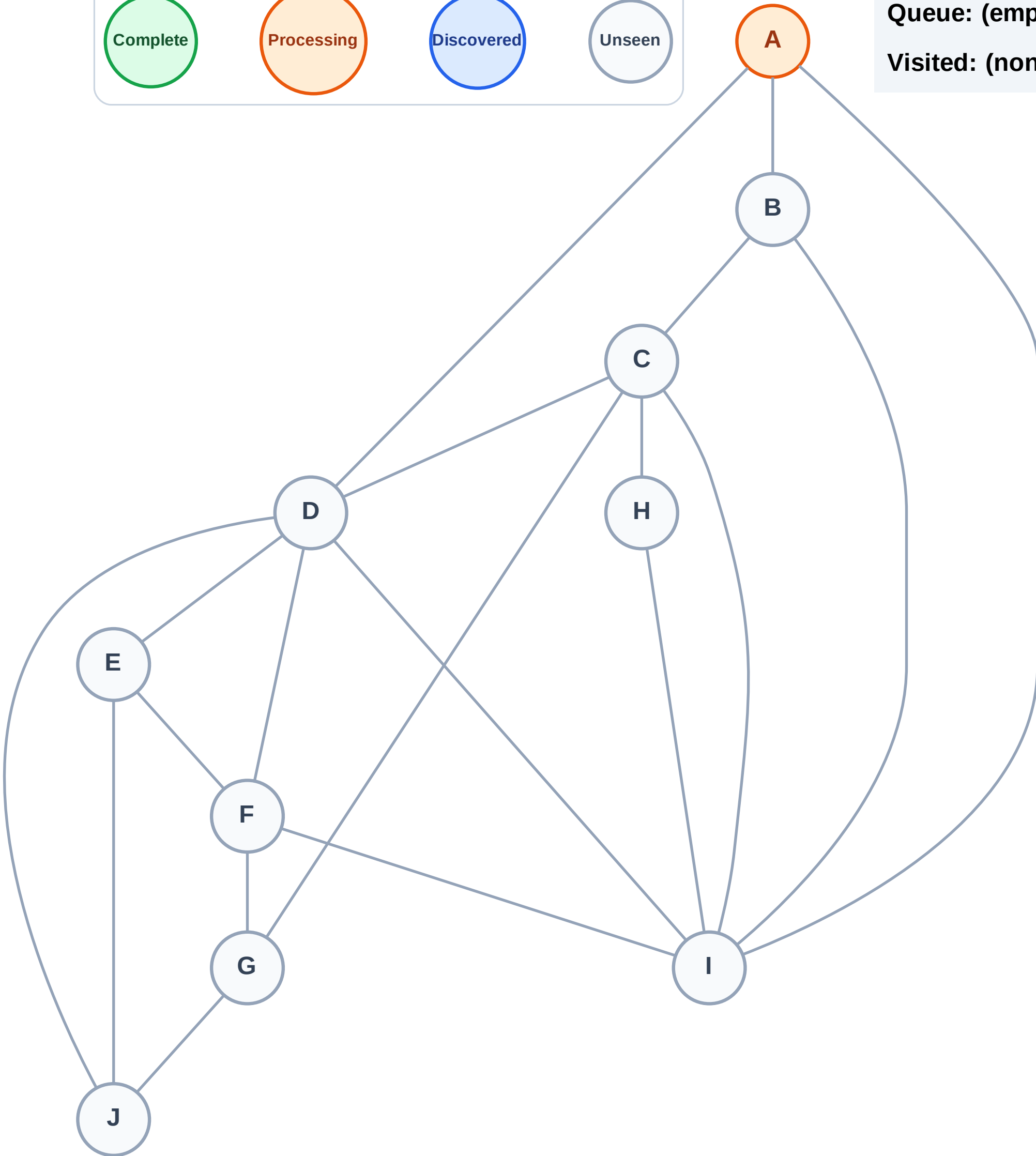
Complete

Processing

Discovered

Unseen

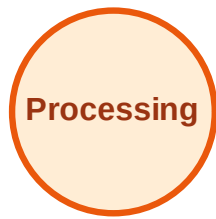
Queue: (empty)
Visited: (none)



BFS Traversal

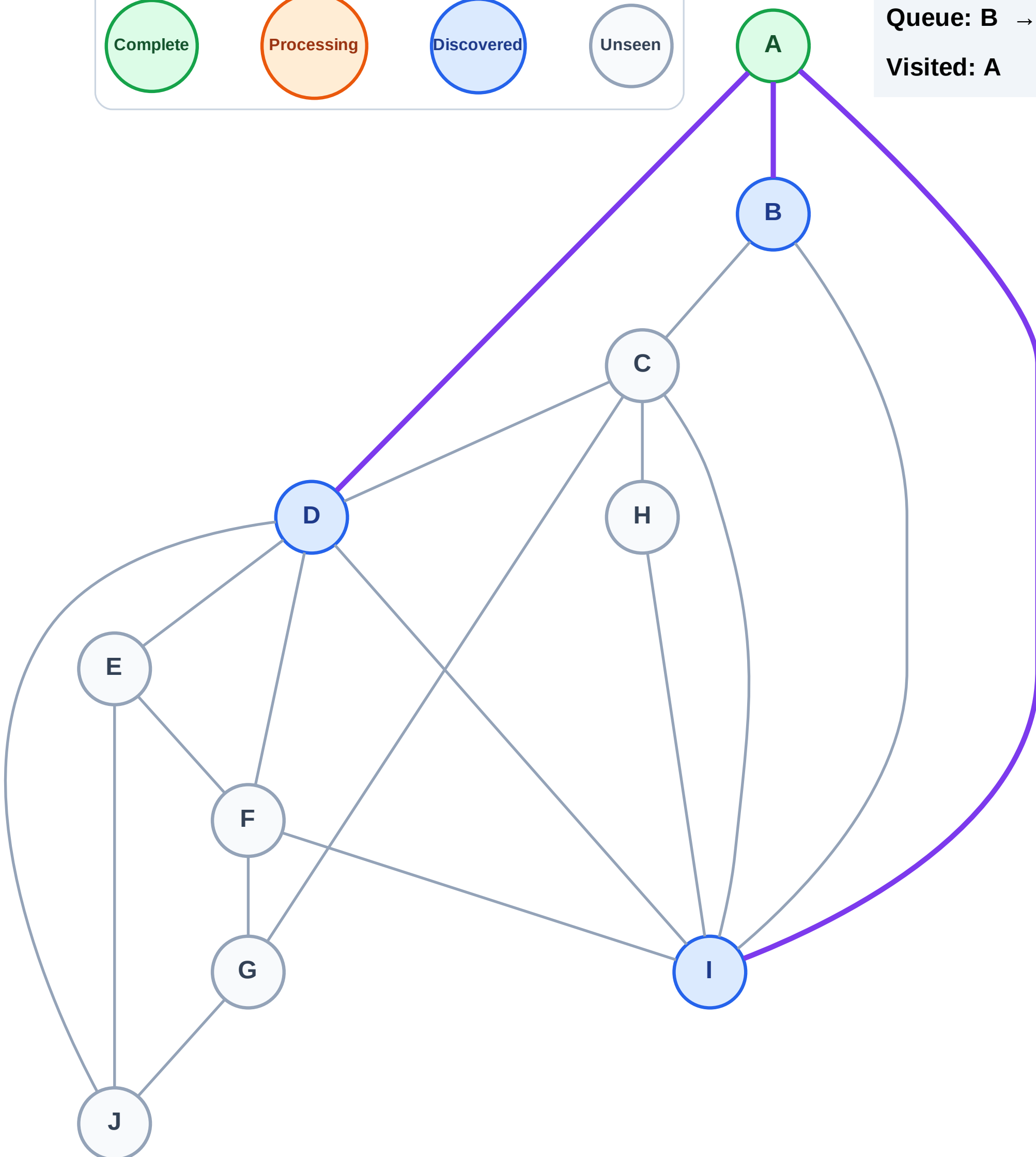
Step 3: Discover B, D, I from A

Legend



Queue: B → D → I

Visited: A



BFS Traversal

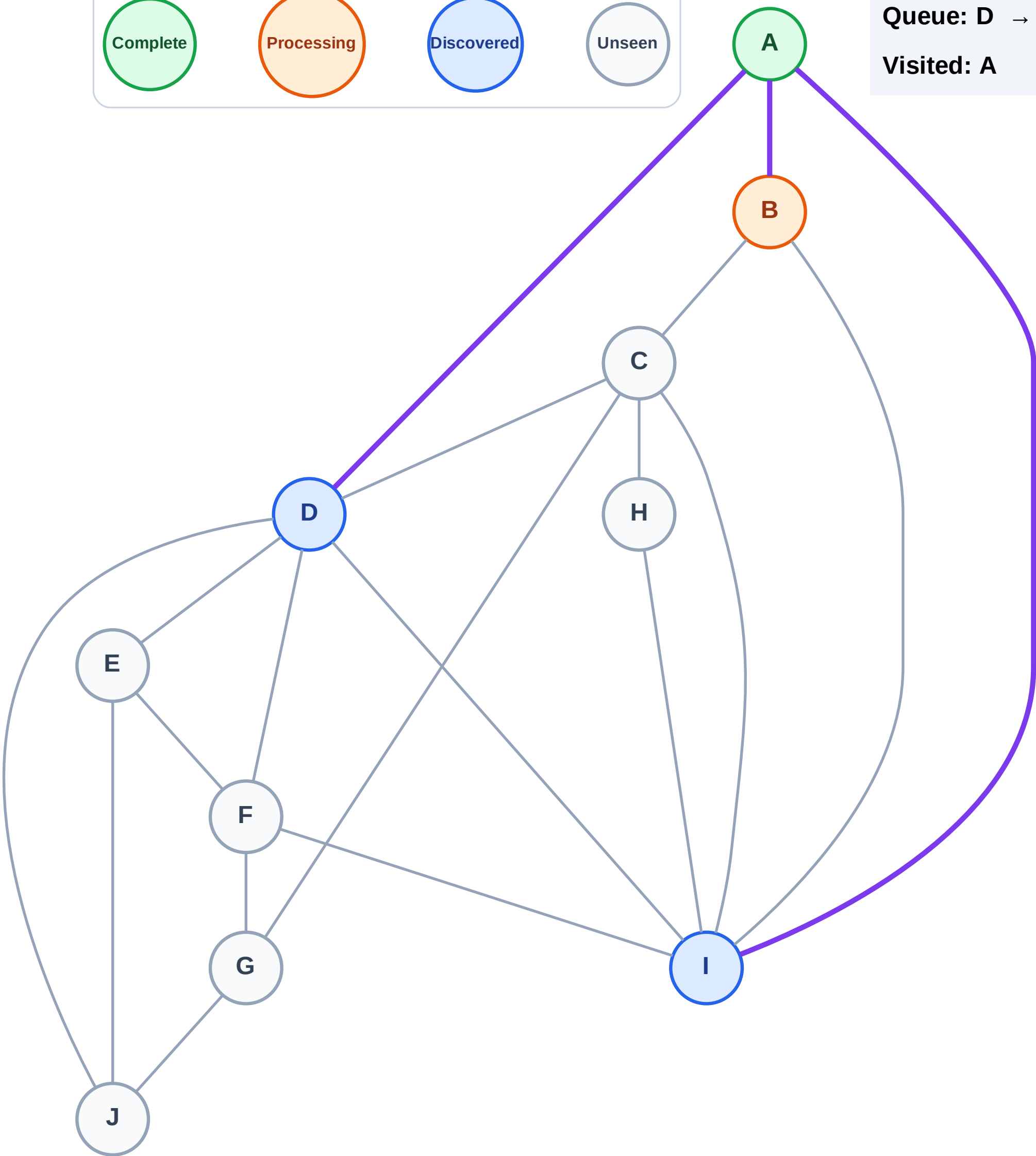
Step 4: Process node B

Legend



Queue: D → I

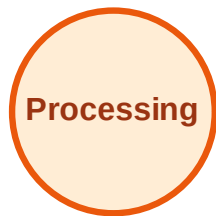
Visited: A



BFS Traversal

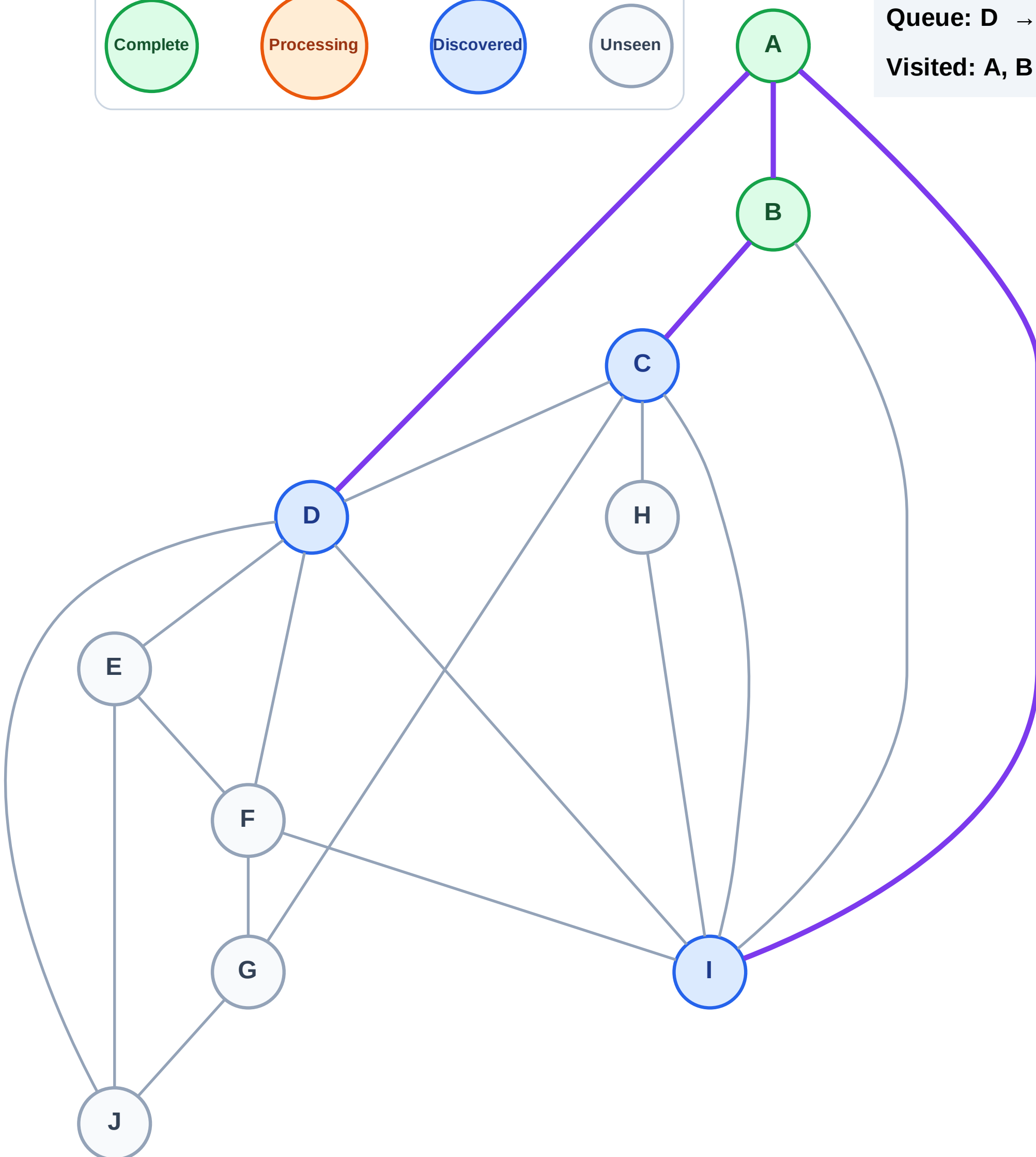
Step 5: Discover C from B

Legend



Queue: D → I → C

Visited: A, B



BFS Traversal

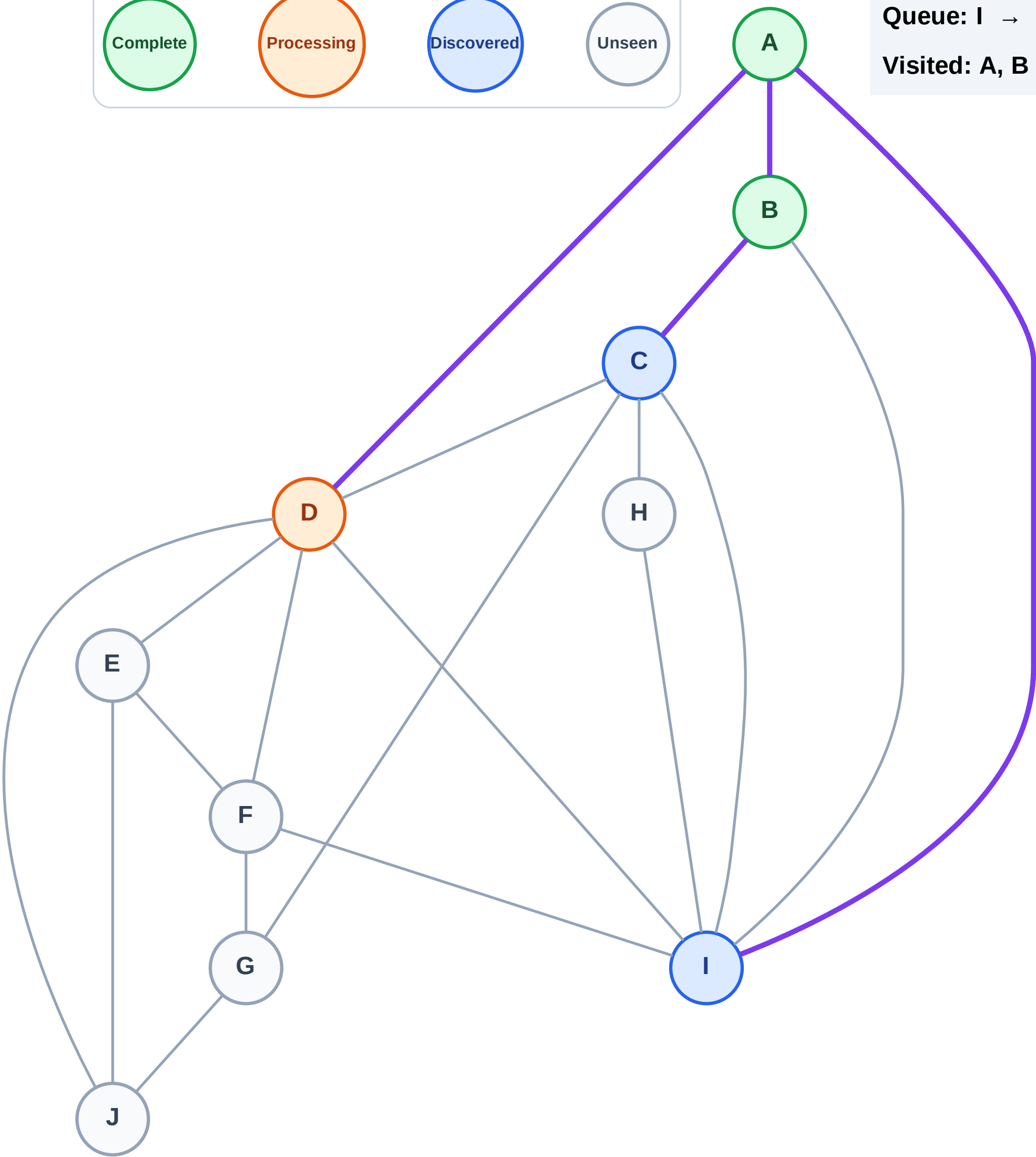
Step 6: Process node D

Legend



Queue: I → C

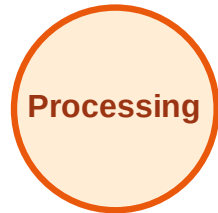
Visited: A, B



BFS Traversal

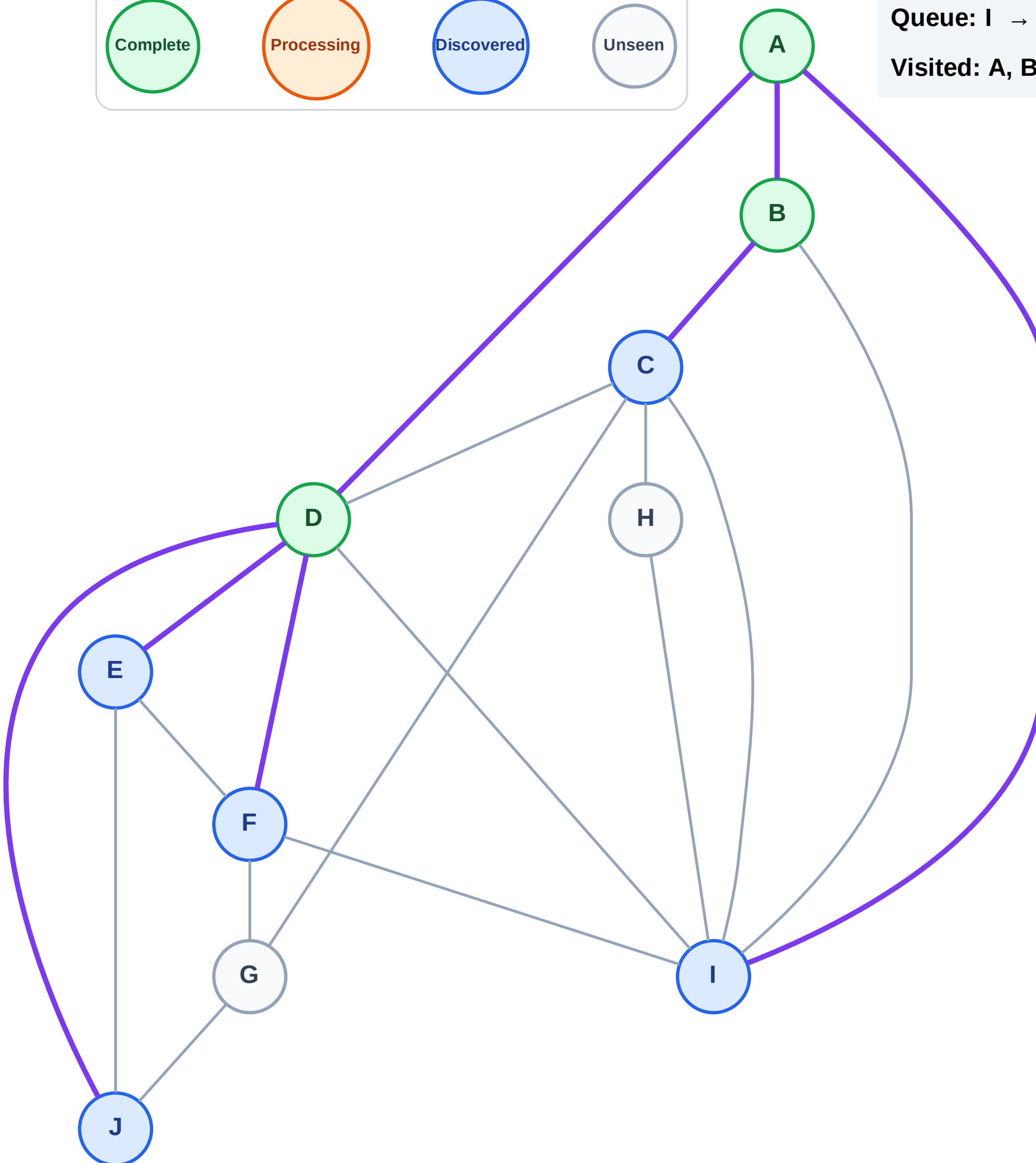
Step 7: Discover E, F, J from D

Legend



Queue: I → C → E → F → J

Visited: A, B, D



BFS Traversal

Step 8: Process node I

Legend

Complete

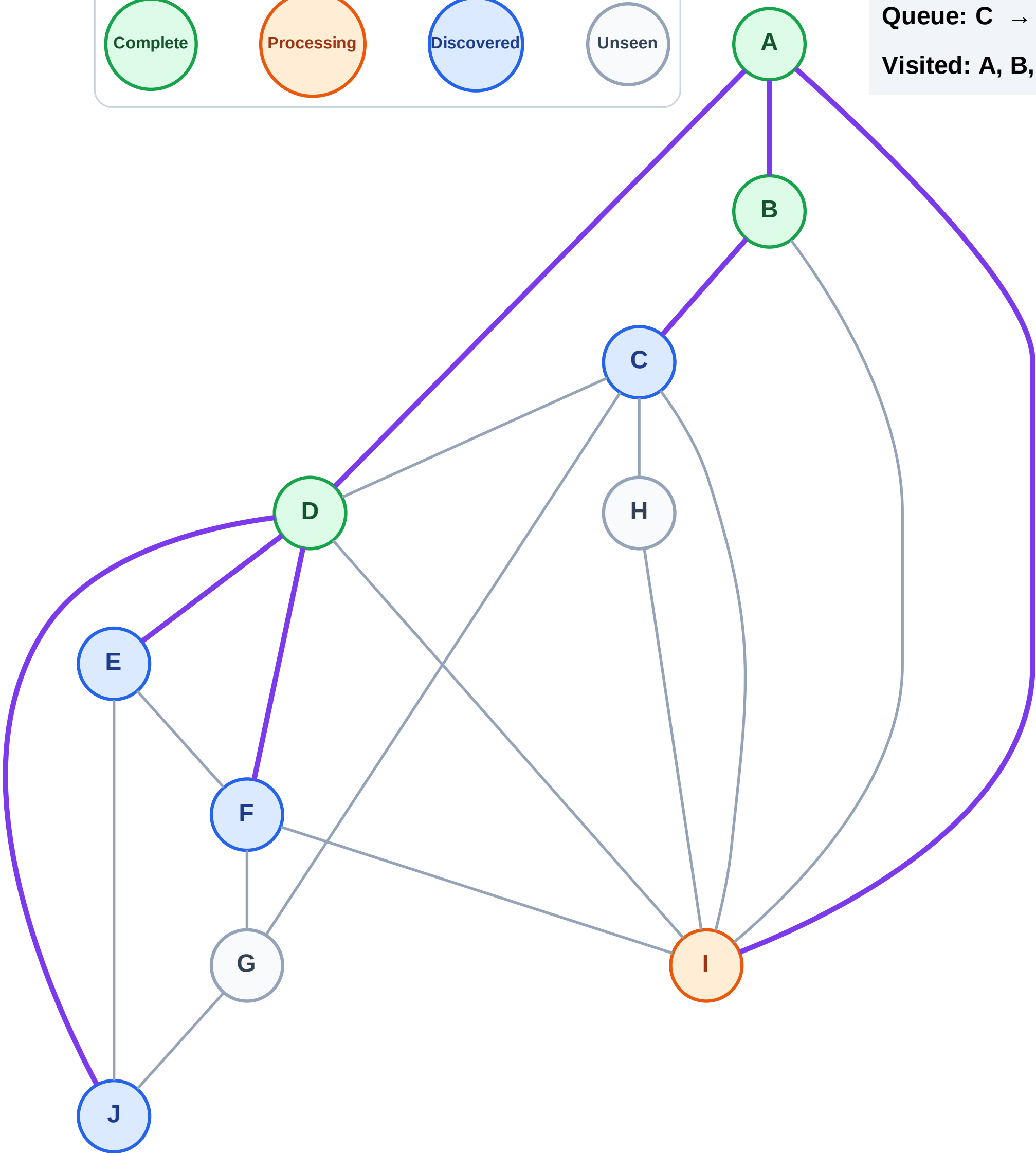
Processing

Discovered

Unseen

Queue: C → E → F → J

Visited: A, B, D



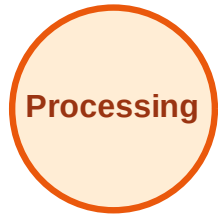
BFS Traversal

Step 9: Discover H from I

Legend



Complete



Processing



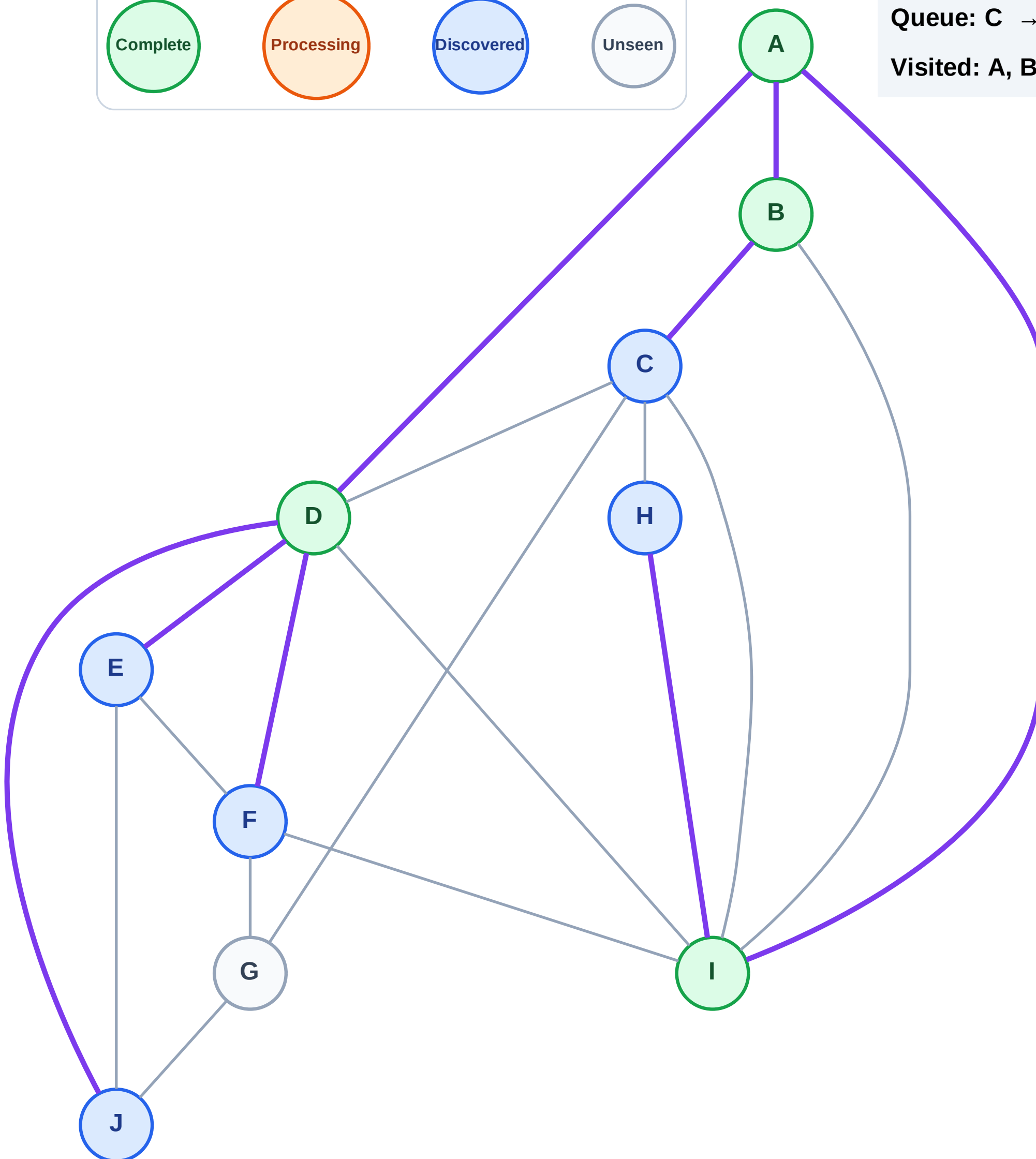
Discovered



Unseen

Queue: C → E → F → J → H

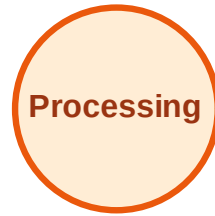
Visited: A, B, D, I



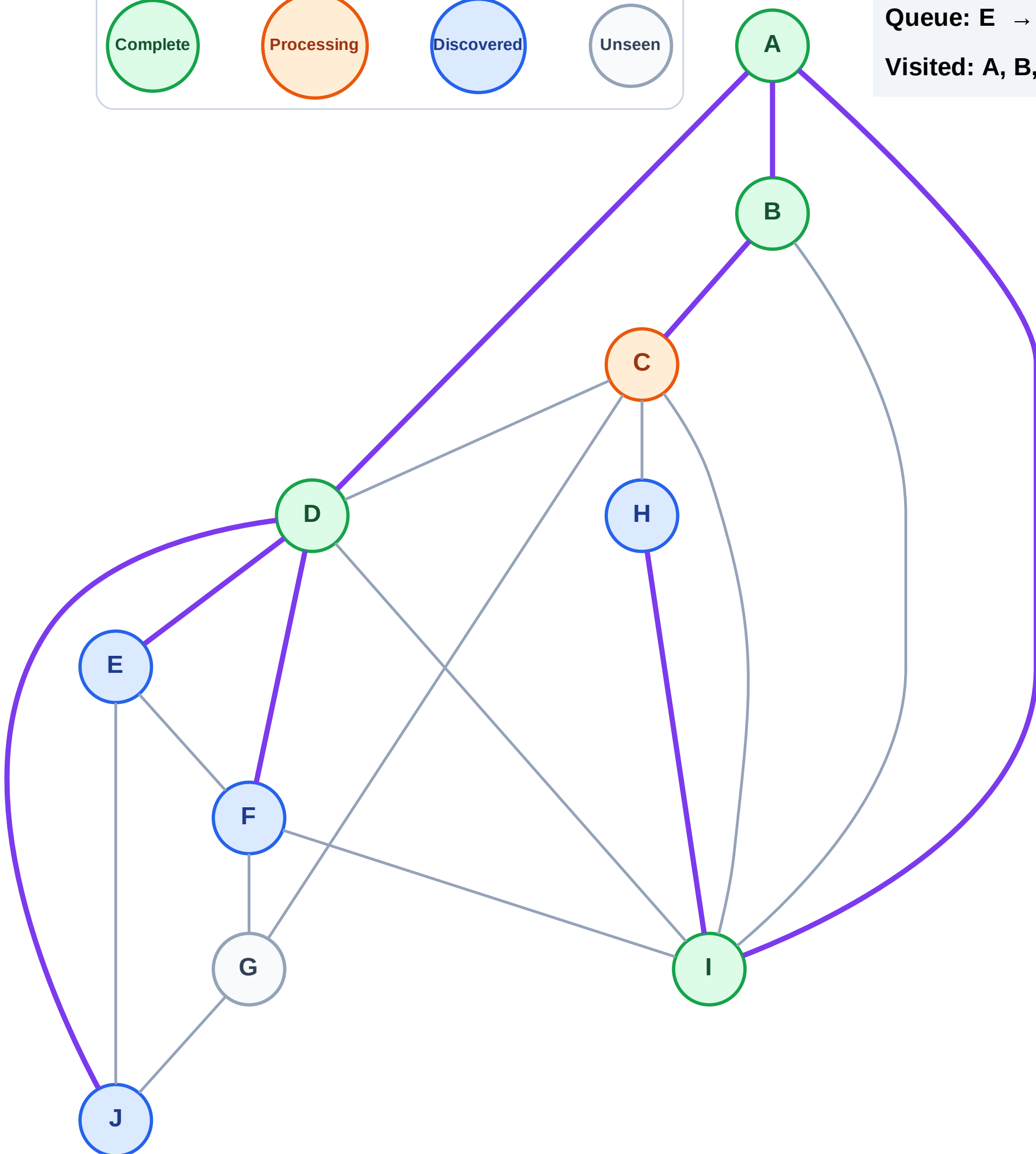
BFS Traversal

Step 10: Process node C

Legend



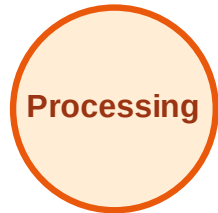
Queue: E → F → J → H
Visited: A, B, D, I



BFS Traversal

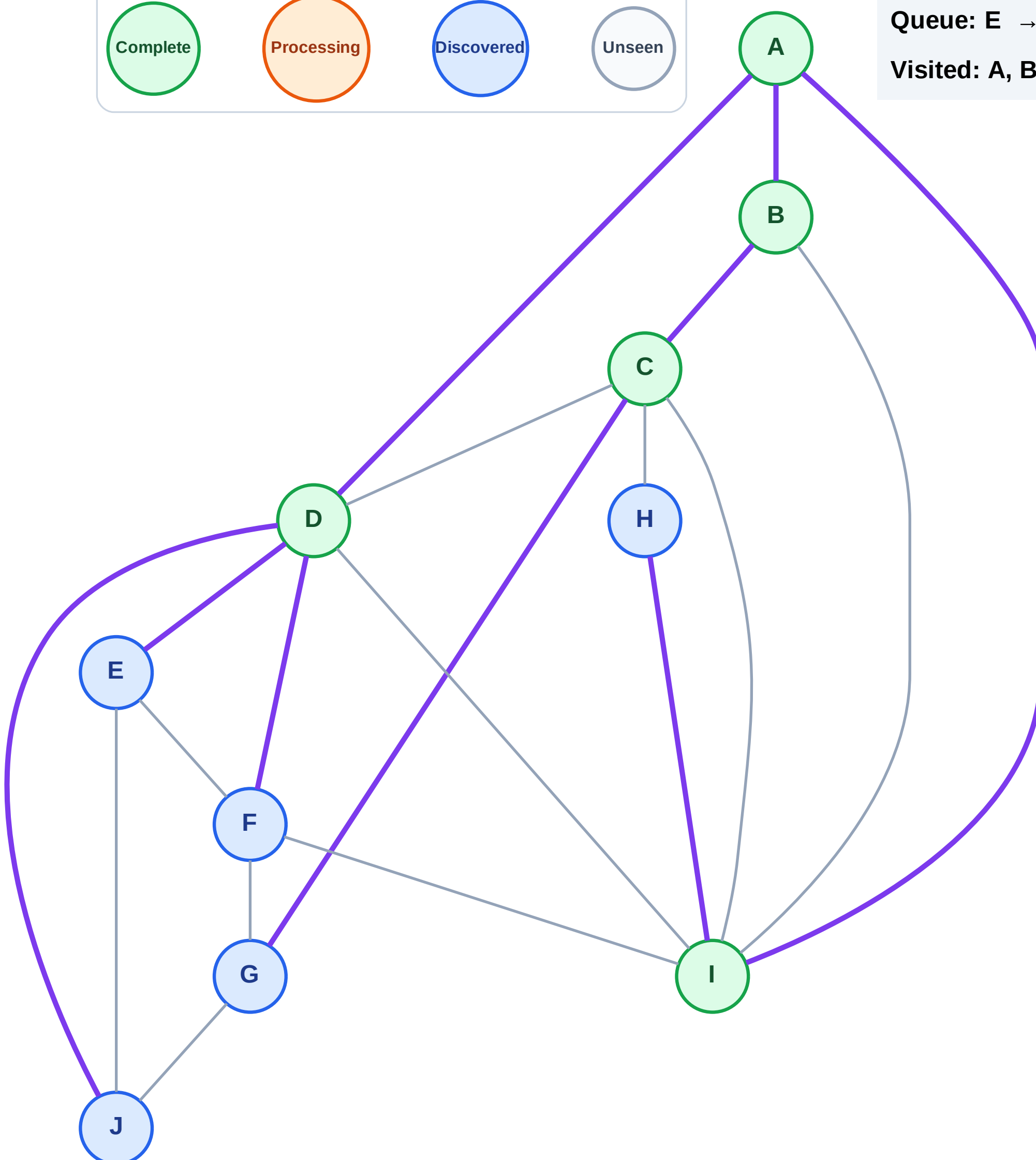
Step 11: Discover G from C

Legend



Queue: E → F → J → H → G

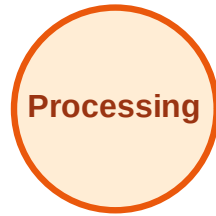
Visited: A, B, C, D, I



BFS Traversal

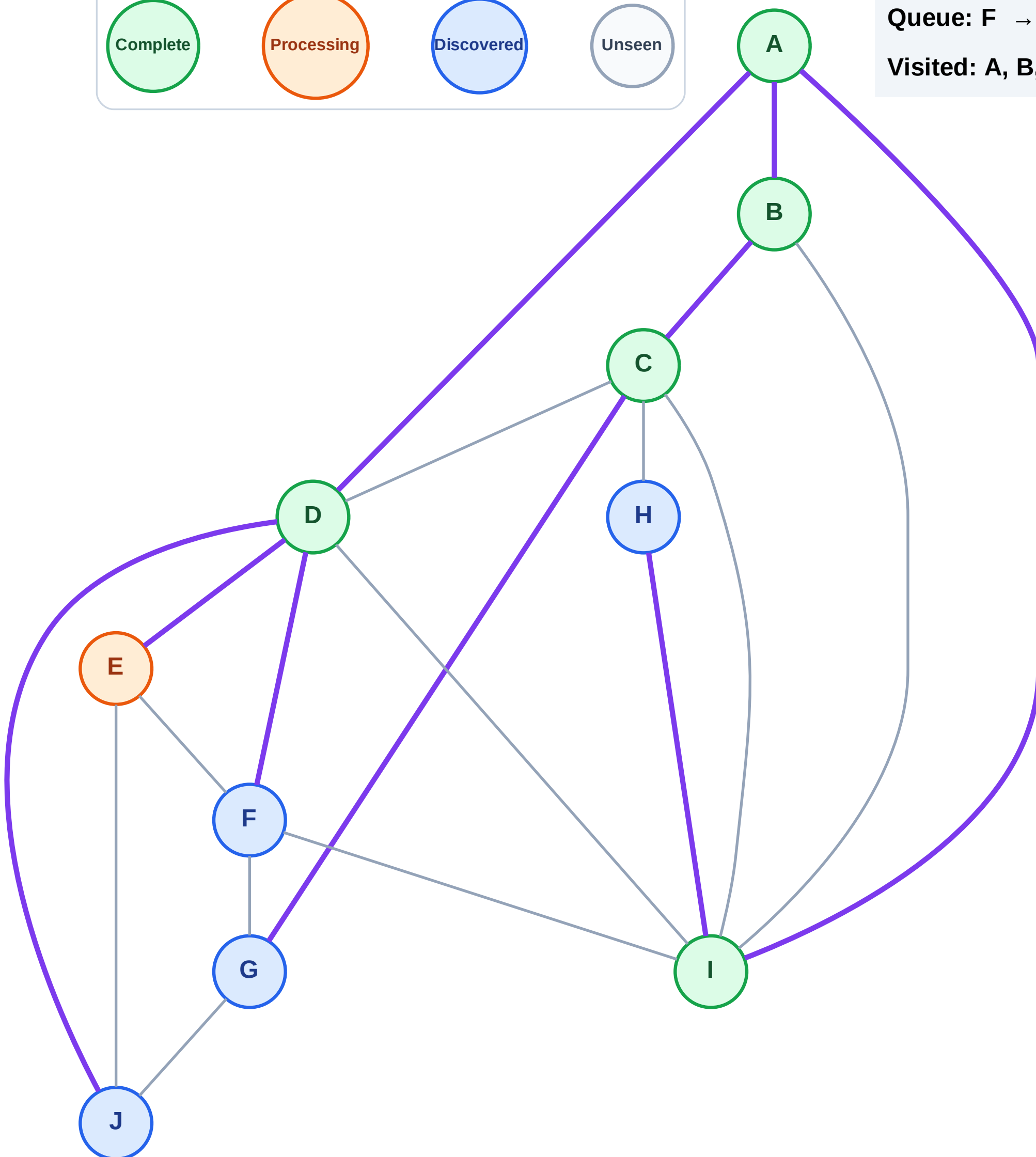
Step 12: Process node E

Legend



Queue: F → J → H → G

Visited: A, B, C, D, I



BFS Traversal

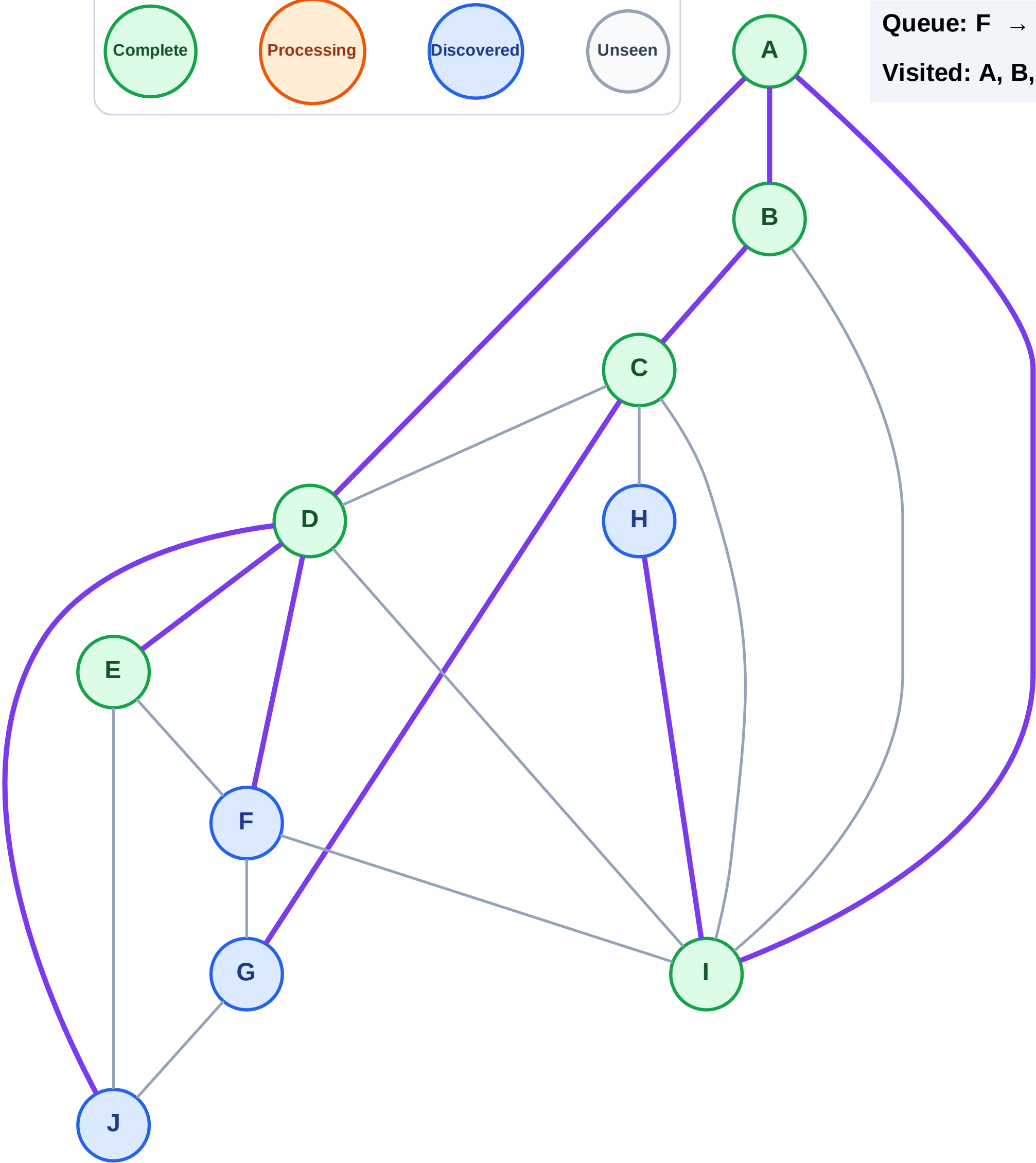
Step 13: No new neighbors from E

Legend



Queue: F → J → H → G

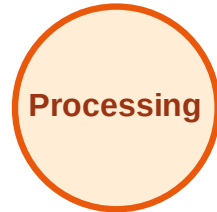
Visited: A, B, C, D, E, I



BFS Traversal

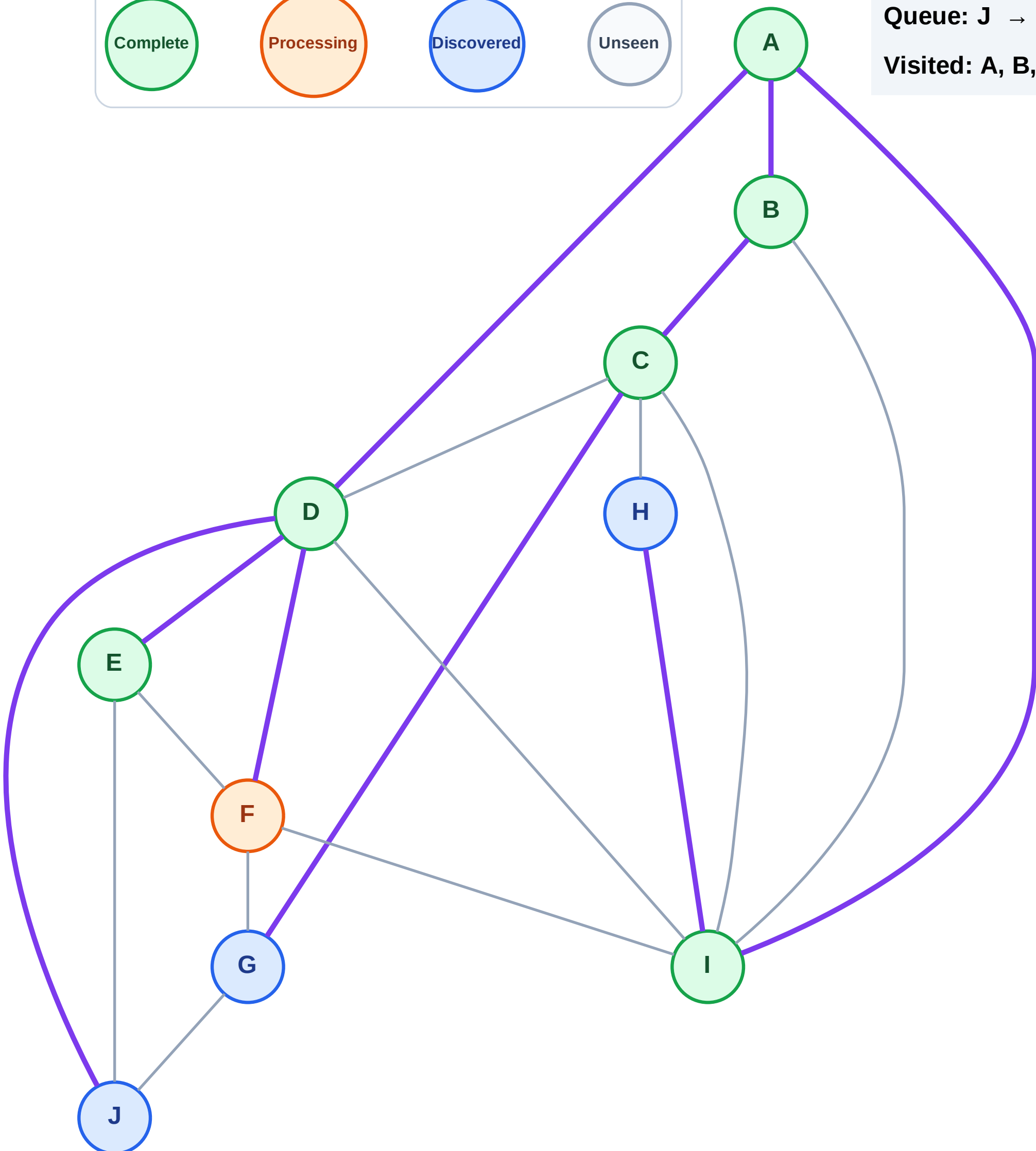
Step 14: Process node F

Legend



Queue: J → H → G

Visited: A, B, C, D, E, I



BFS Traversal

Step 15: No new neighbors from F

Legend

Complete

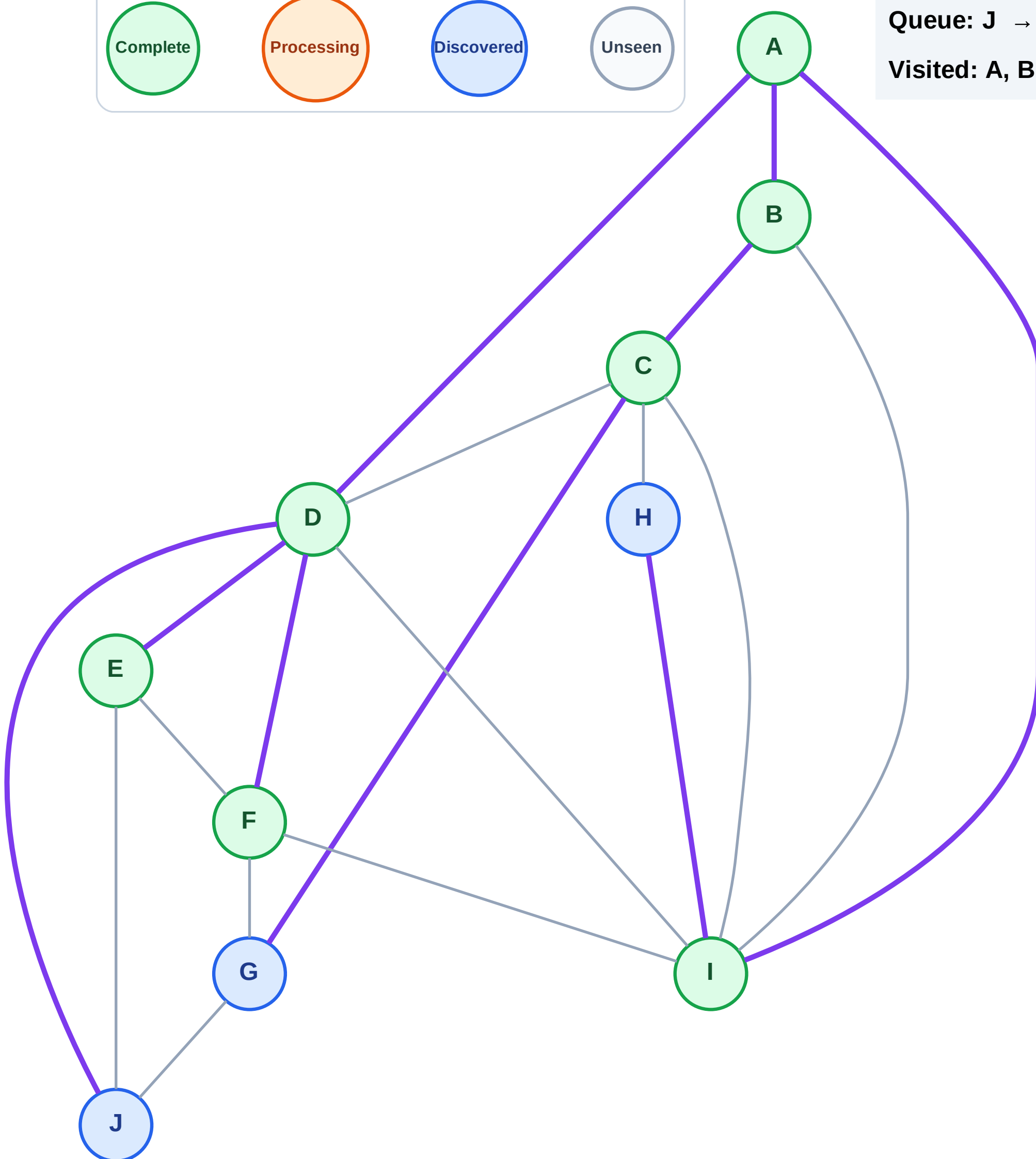
Processing

Discovered

Unseen

Queue: J → H → G

Visited: A, B, C, D, E, F, I



BFS Traversal

Step 16: Process node J

Legend

Complete

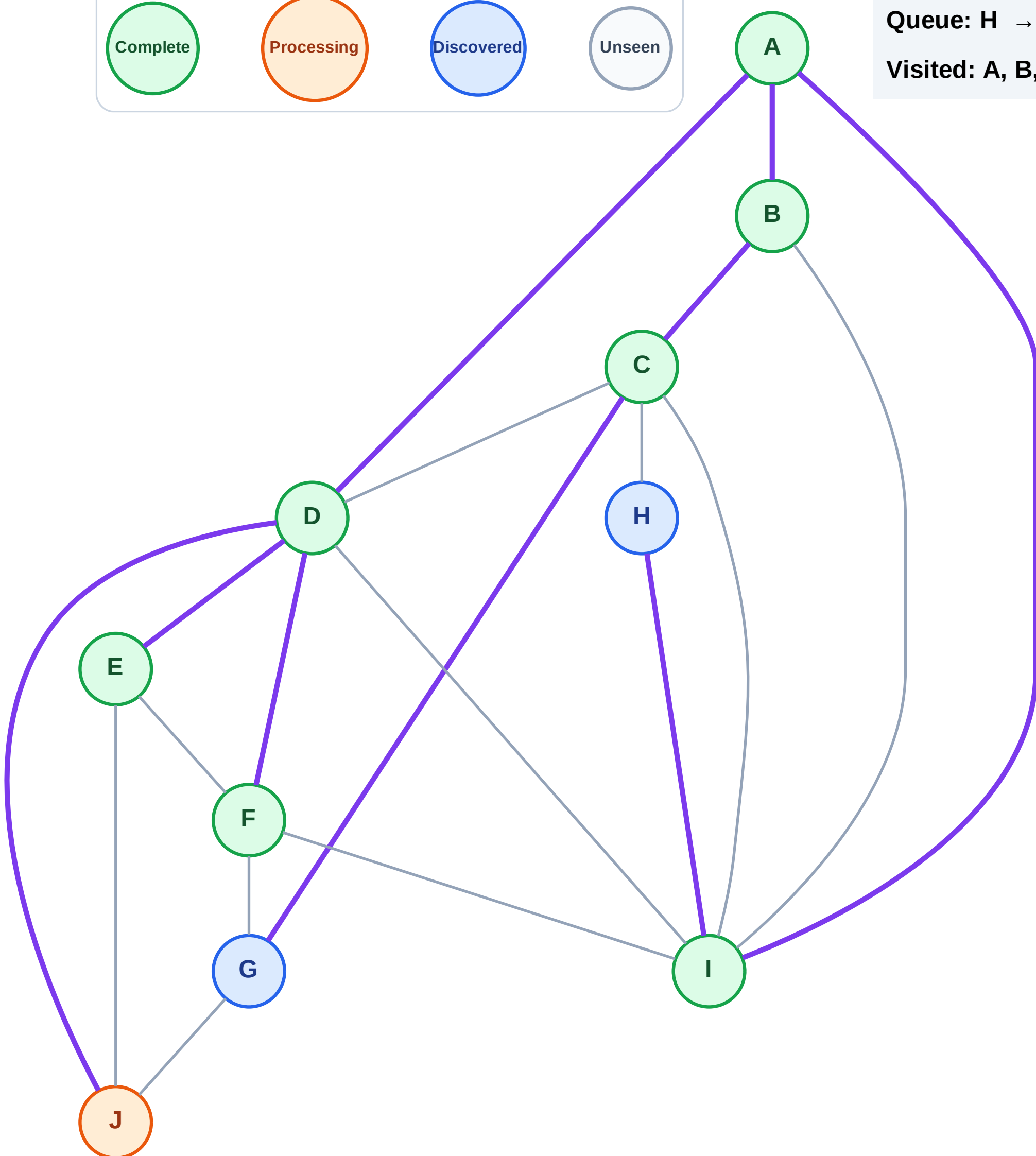
Processing

Discovered

Unseen

Queue: H → G

Visited: A, B, C, D, E, F, I



BFS Traversal

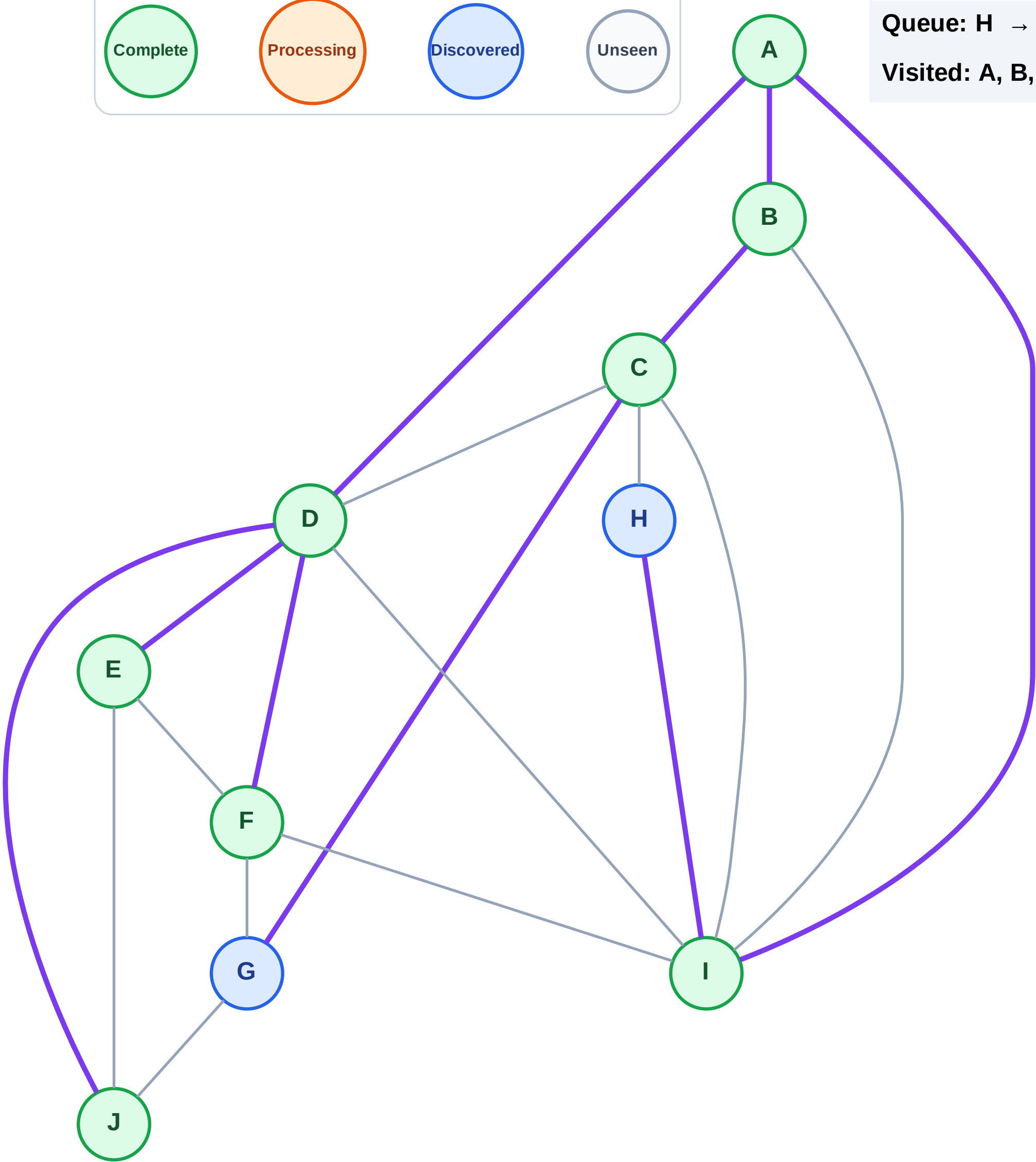
Step 17: No new neighbors from J

Legend



Queue: H → G

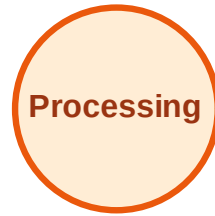
Visited: A, B, C, D, E, F, I, J



BFS Traversal

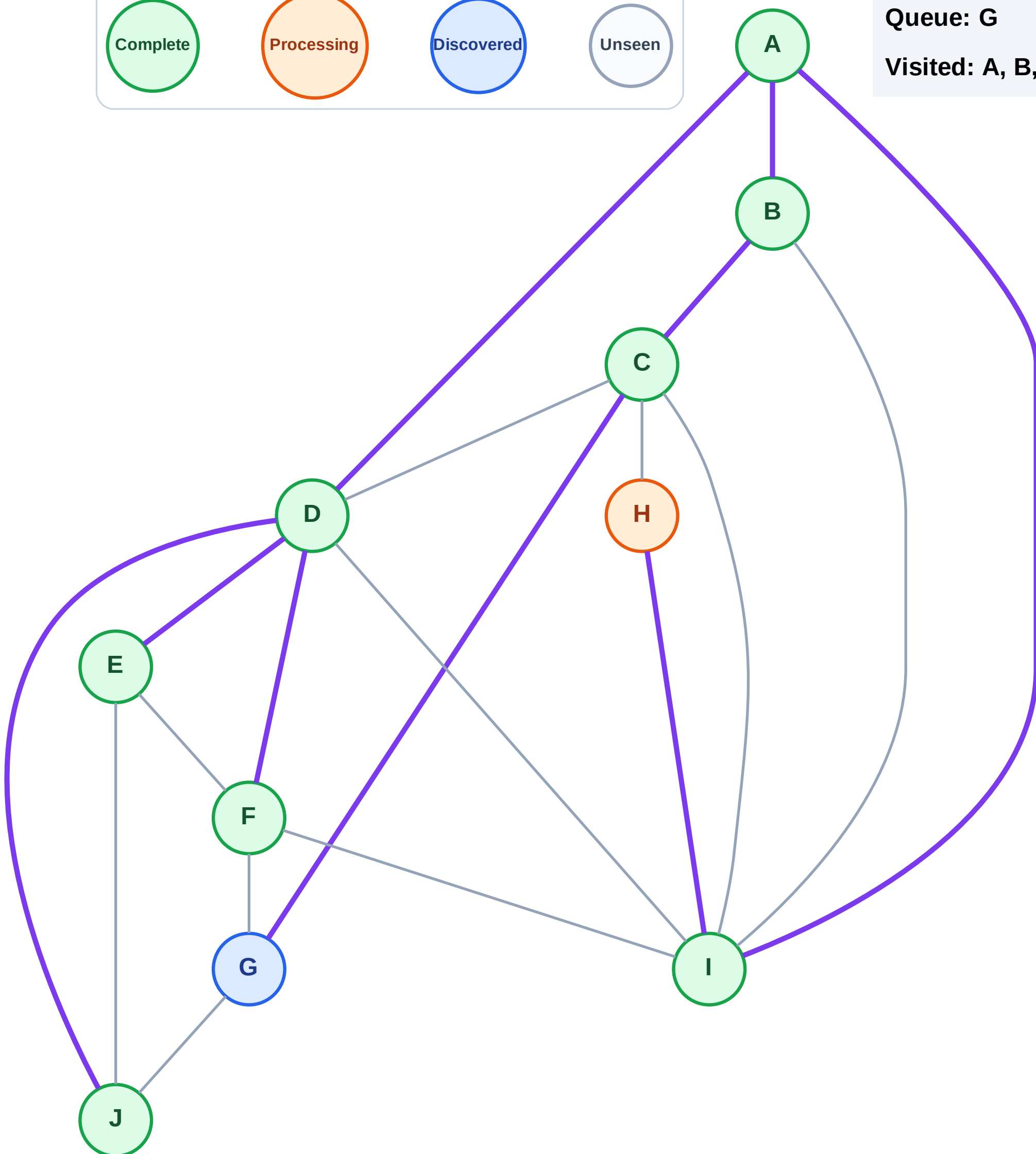
Step 18: Process node H

Legend



Queue: G

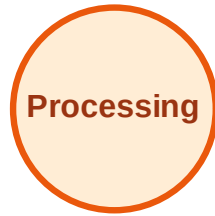
Visited: A, B, C, D, E, F, I, J



BFS Traversal

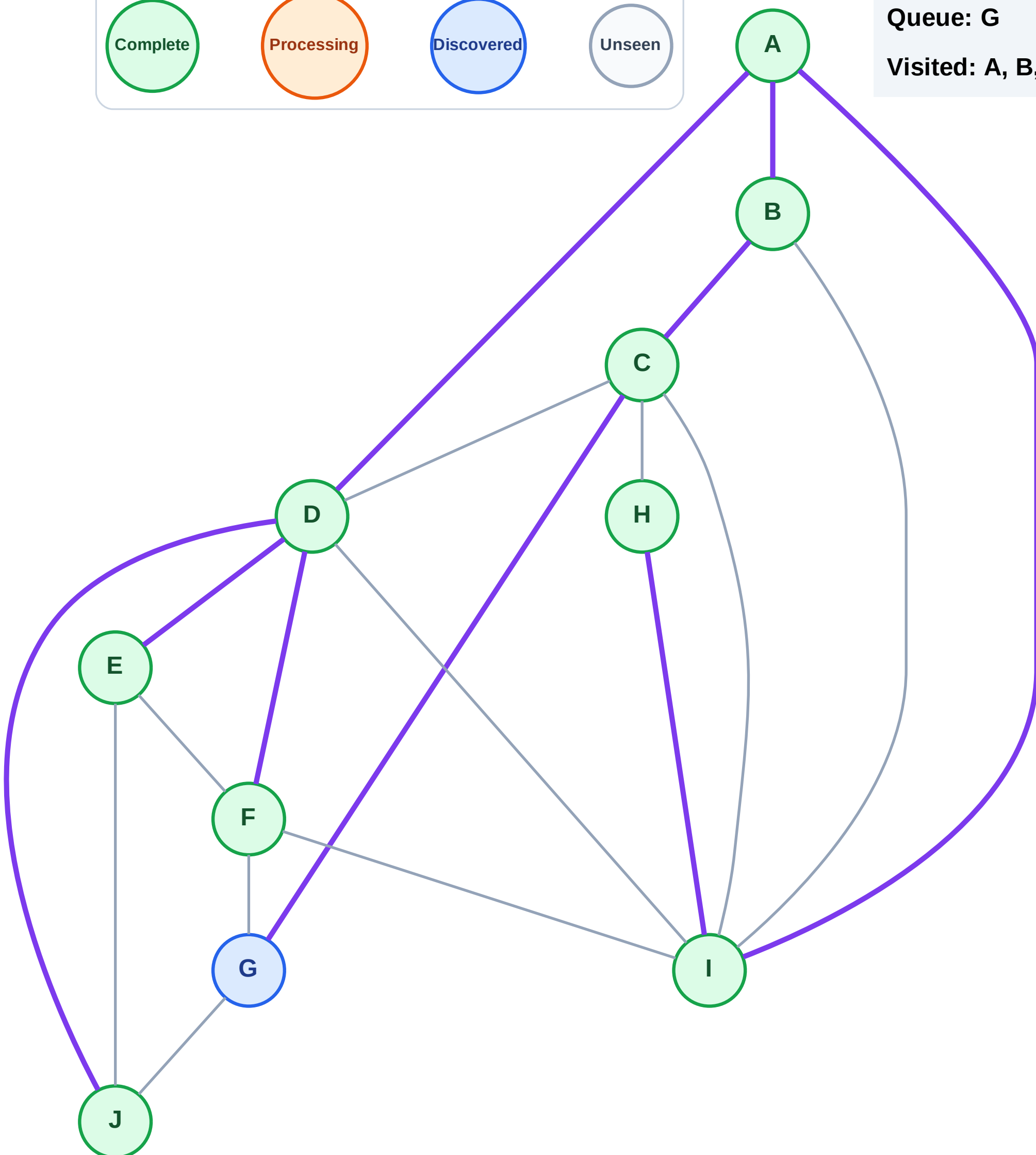
Step 19: No new neighbors from H

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

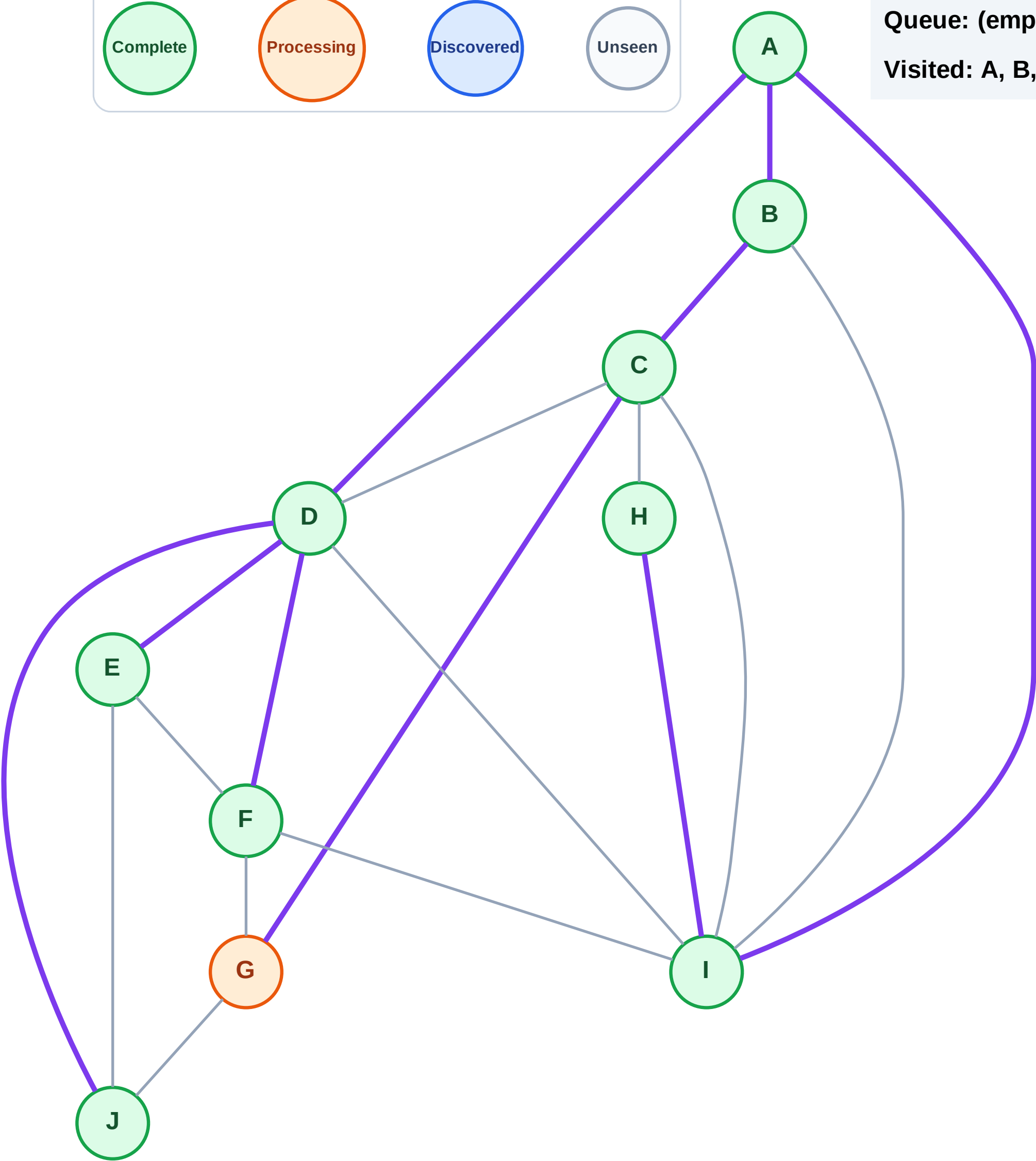
Step 20: Process node G

Legend



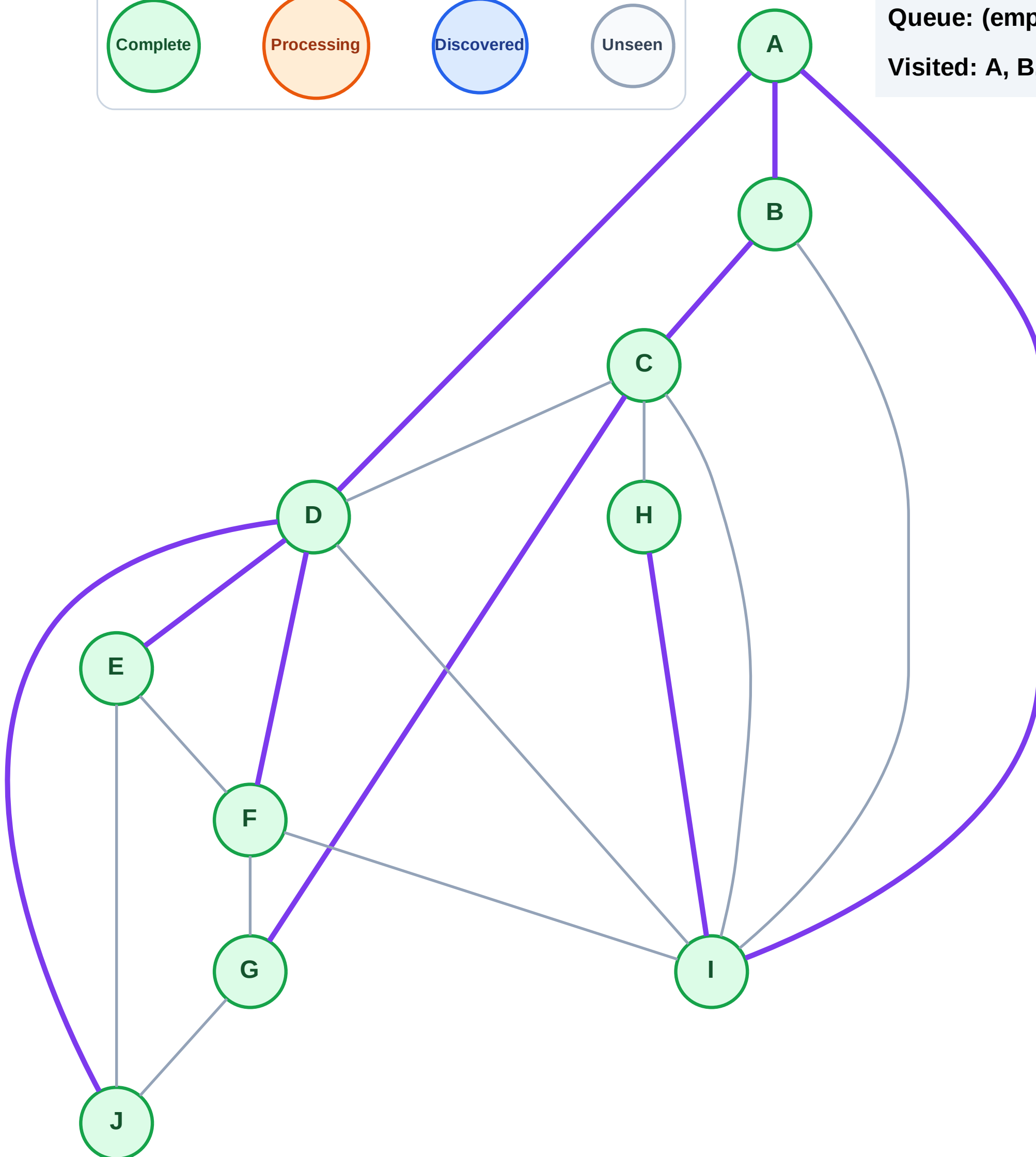
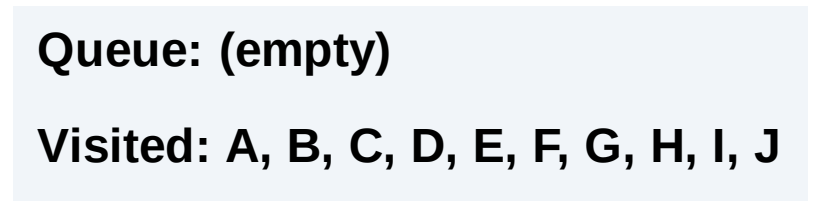
Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



Step 21: No new neighbors from G

Step 21: No new neighbors from G



DFS Traversal

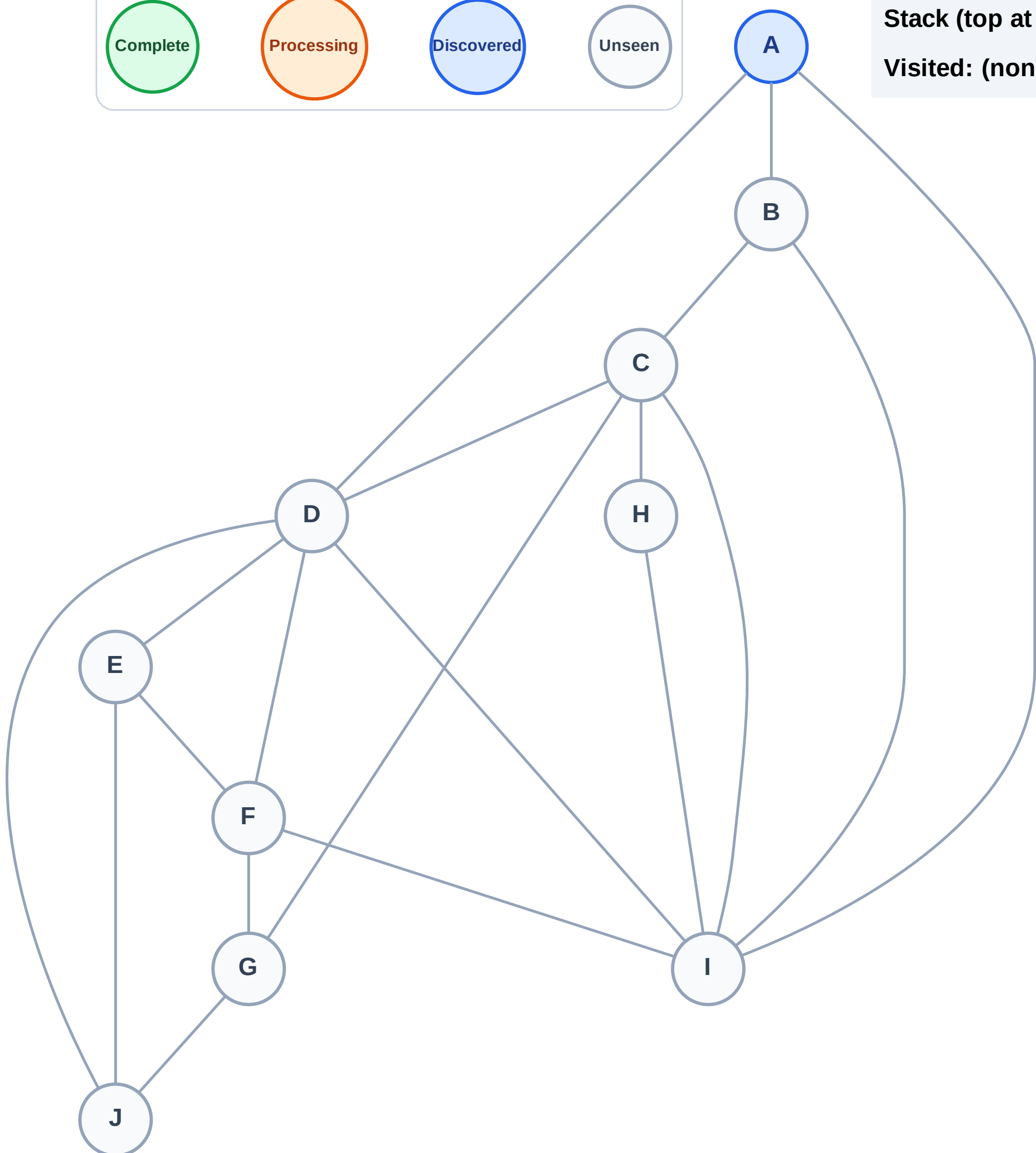
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

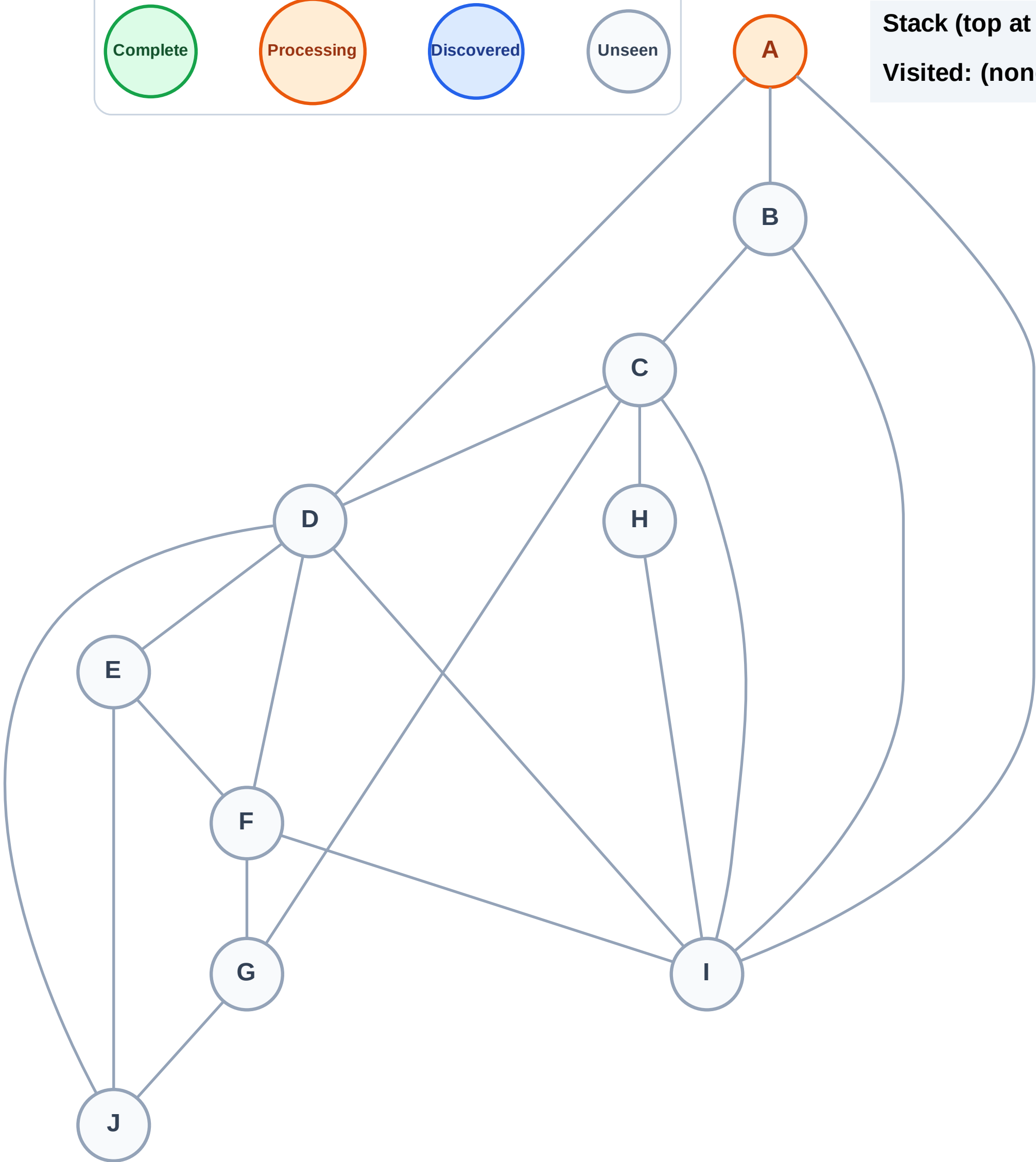
Complete

Processing

Discovered

Unseen

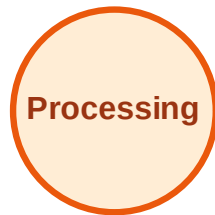
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

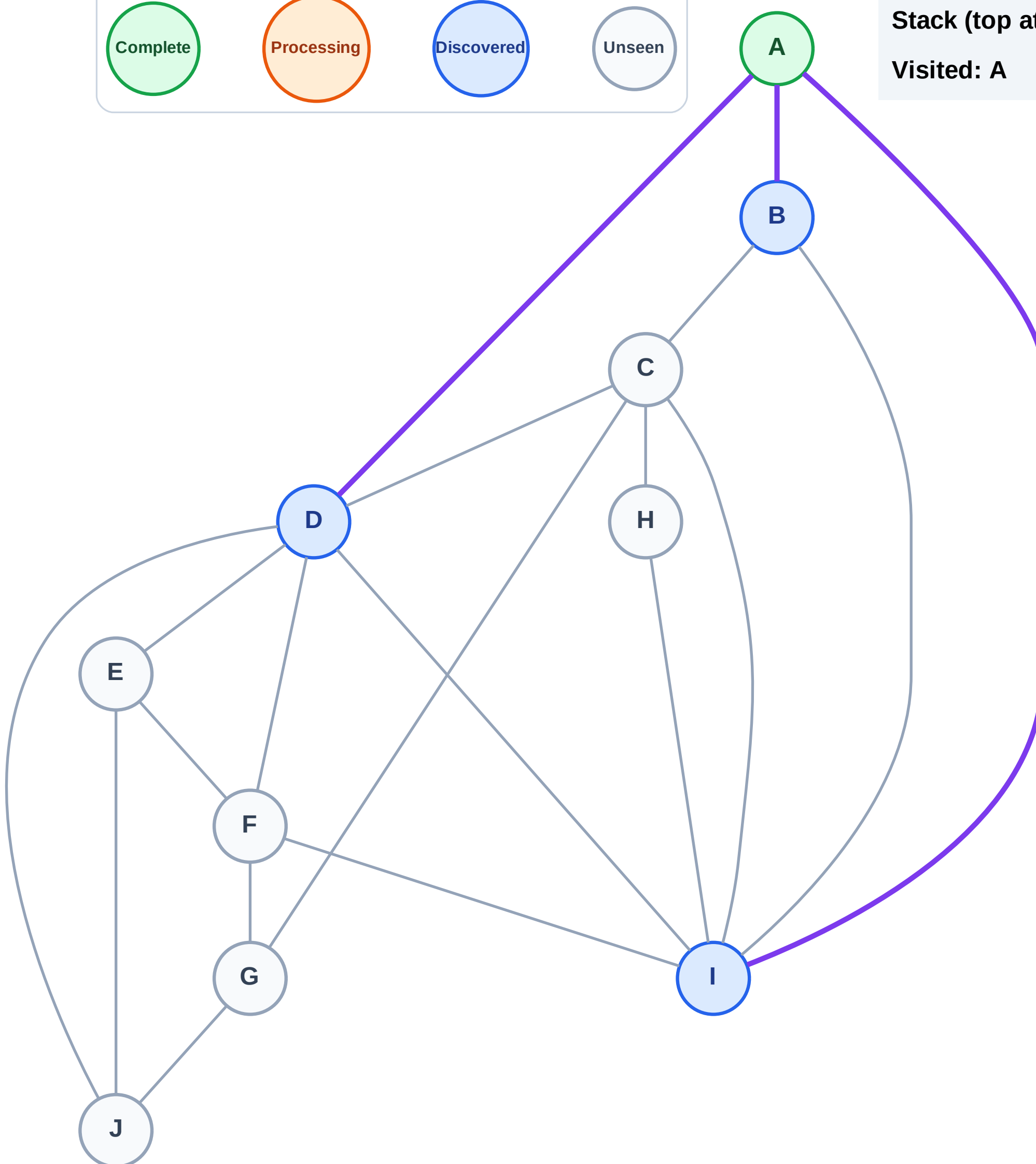
Step 3: Discover I, D, B from A

Legend



Stack (top at right): I | D | B

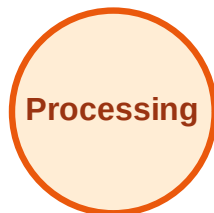
Visited: A



DFS Traversal

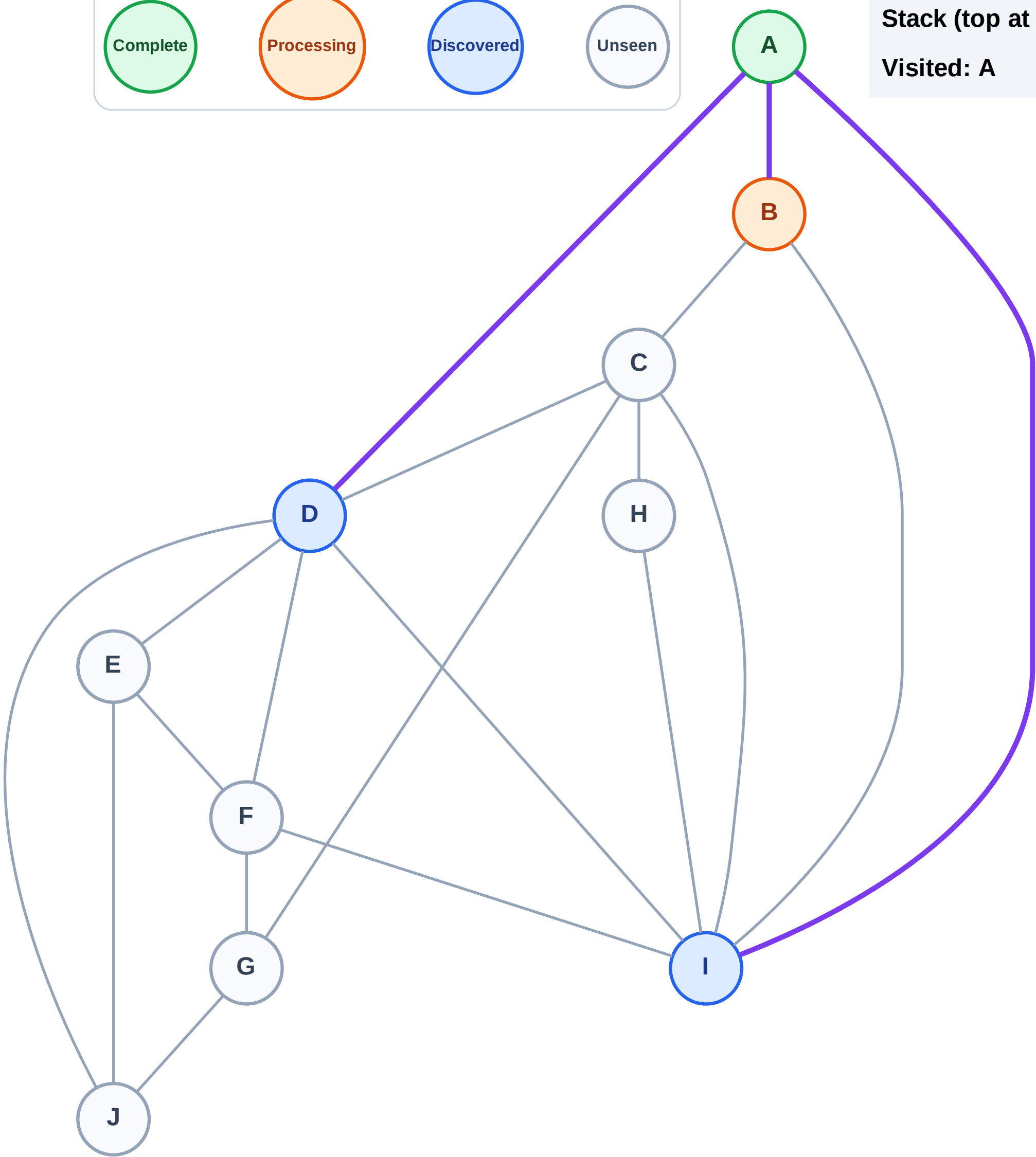
Step 4: Process node B

Legend



Stack (top at right): I | D

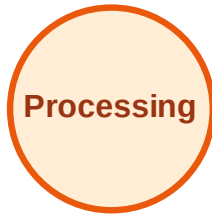
Visited: A



DFS Traversal

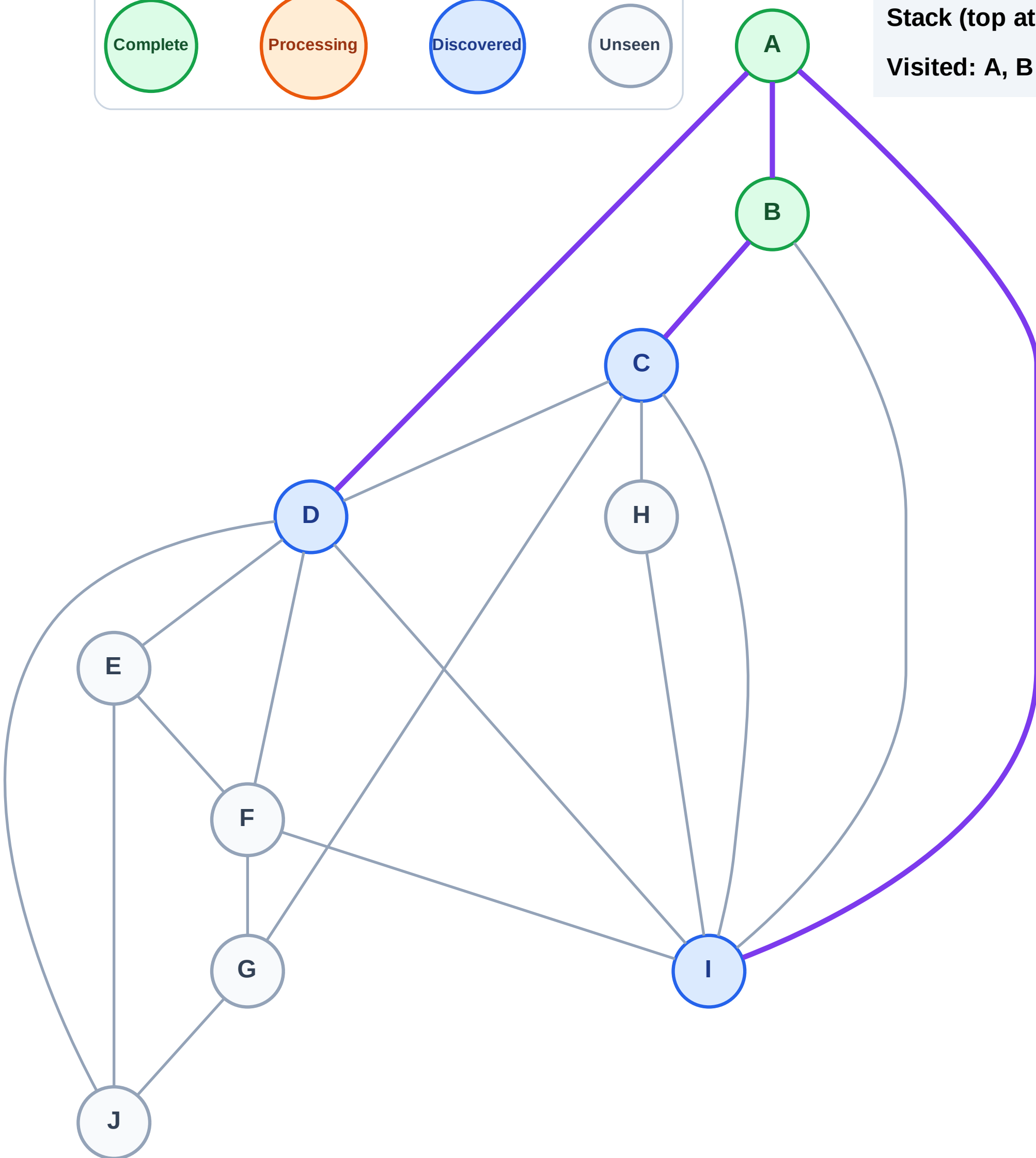
Step 5: Discover C from B

Legend



Stack (top at right): I | D | C

Visited: A, B



DFS Traversal

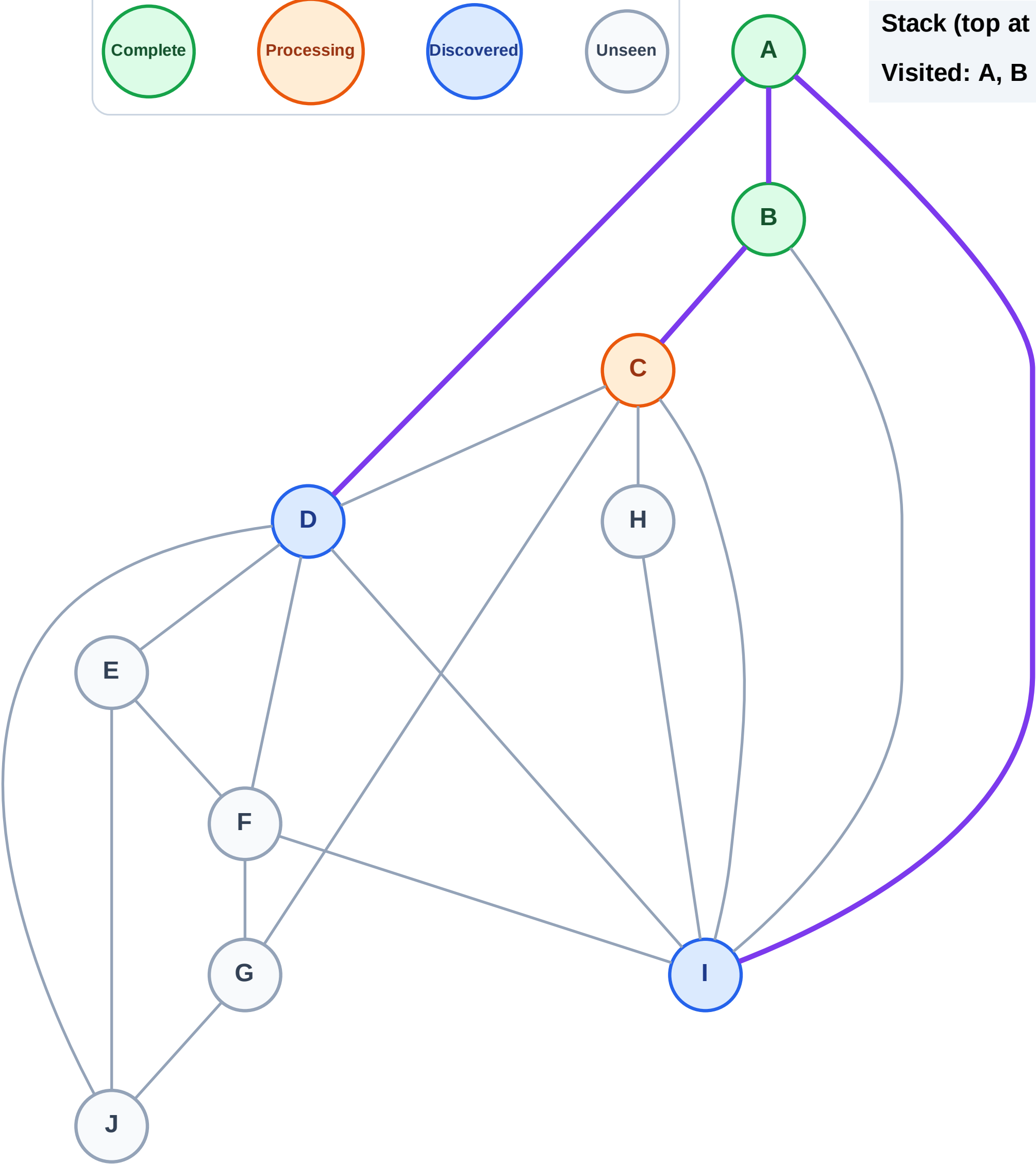
Step 6: Process node C

Legend



Stack (top at right): I | D

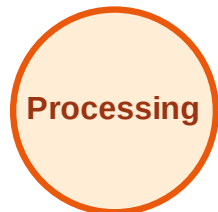
Visited: A, B



DFS Traversal

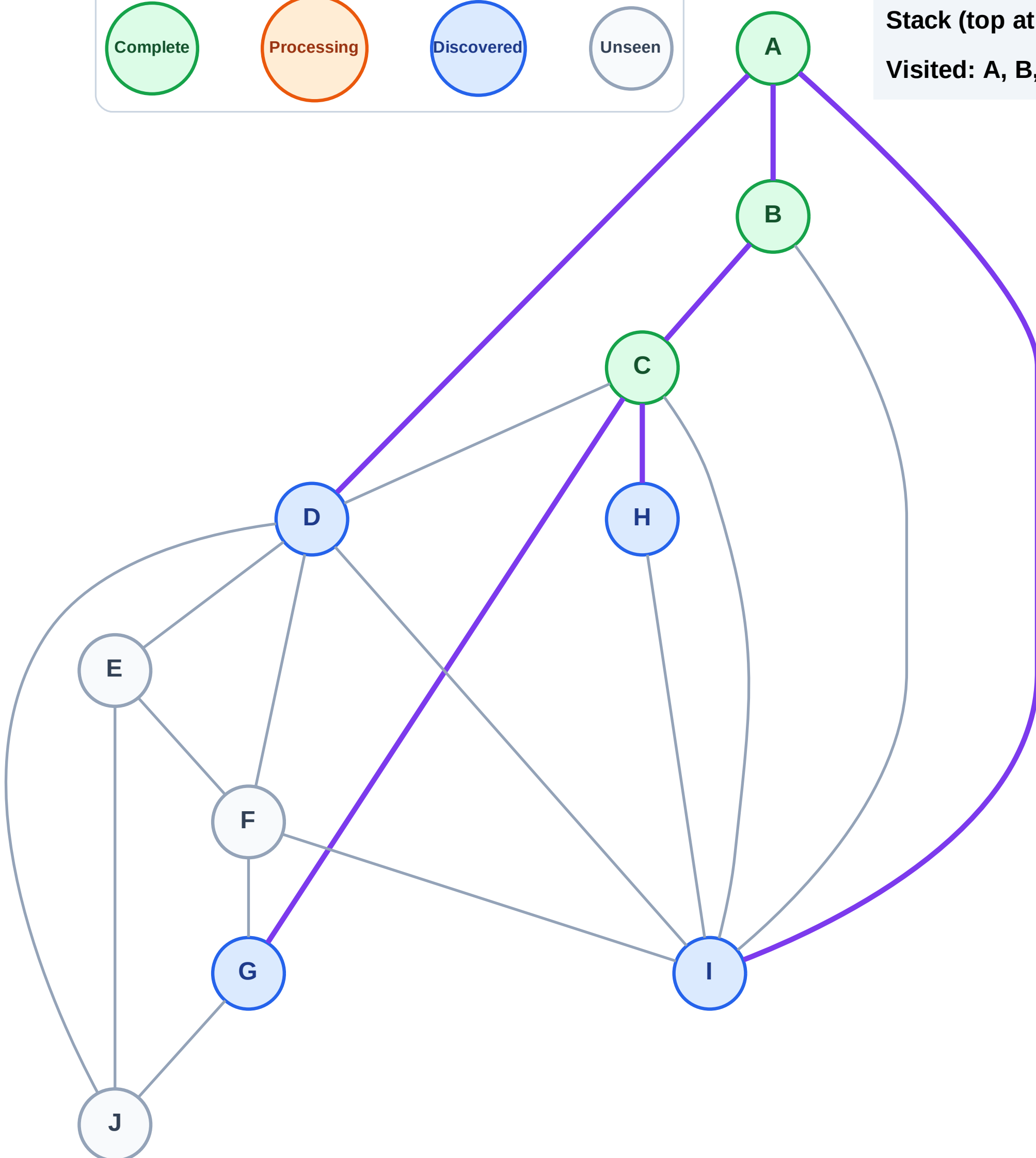
Step 7: Discover H, G from C

Legend



Stack (top at right): I | D | H | G

Visited: A, B, C



DFS Traversal

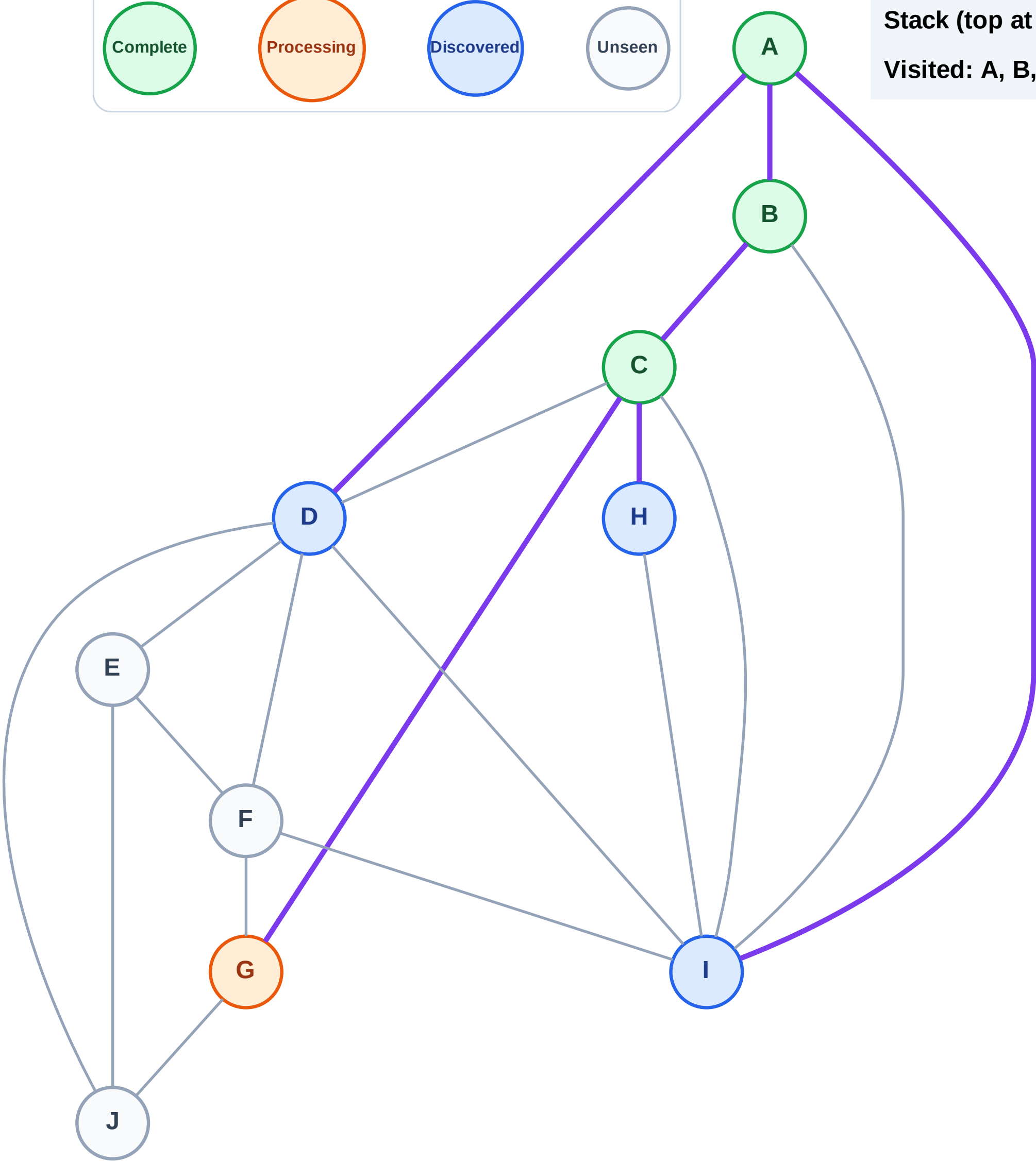
Step 8: Process node G

Legend



Stack (top at right): I | D | H

Visited: A, B, C



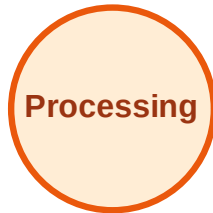
DFS Traversal

Step 9: Discover J, F from G

Legend



Complete



Processing



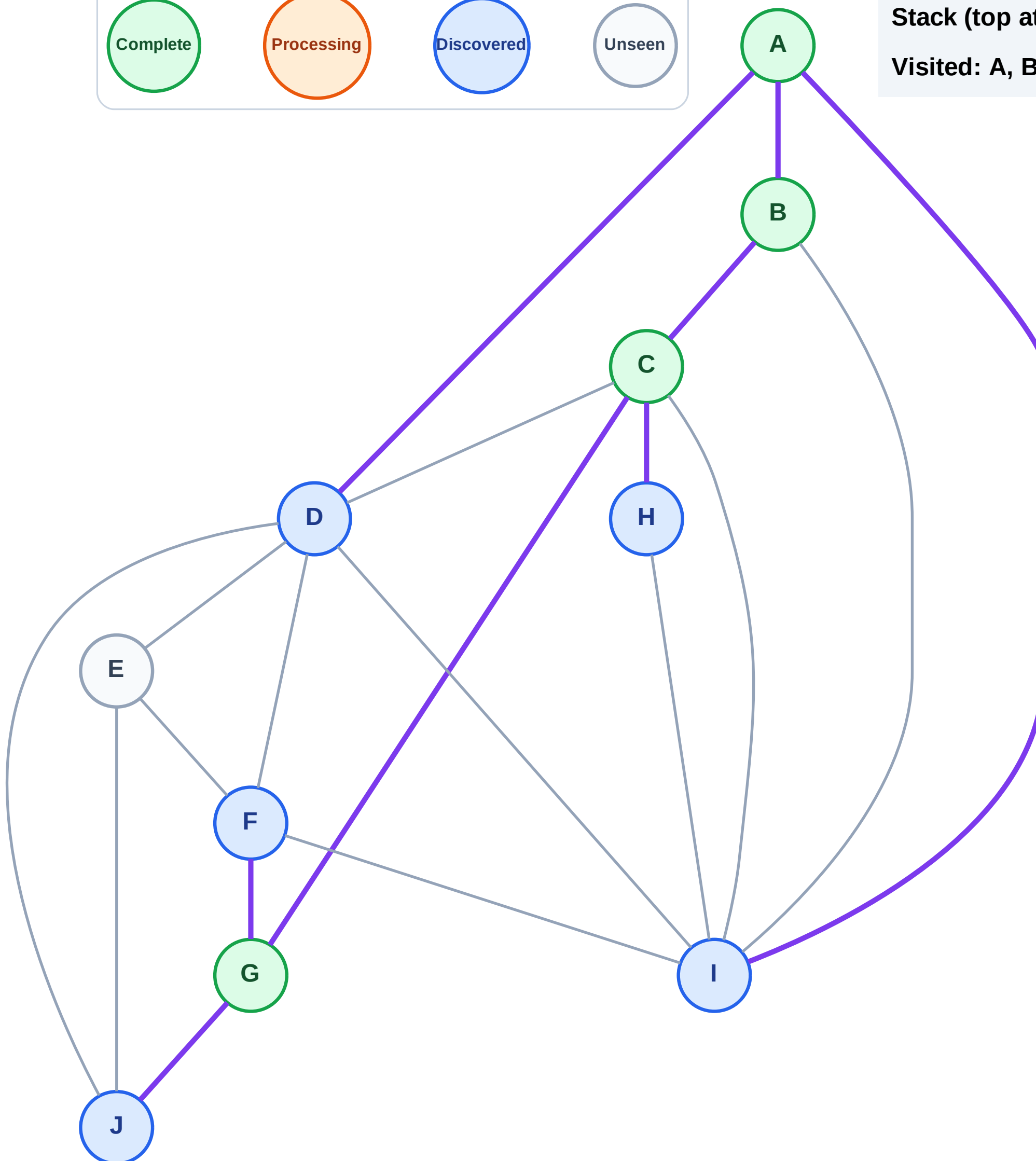
Discovered



Unseen

Stack (top at right): I | D | H | J | F

Visited: A, B, C, G



DFS Traversal

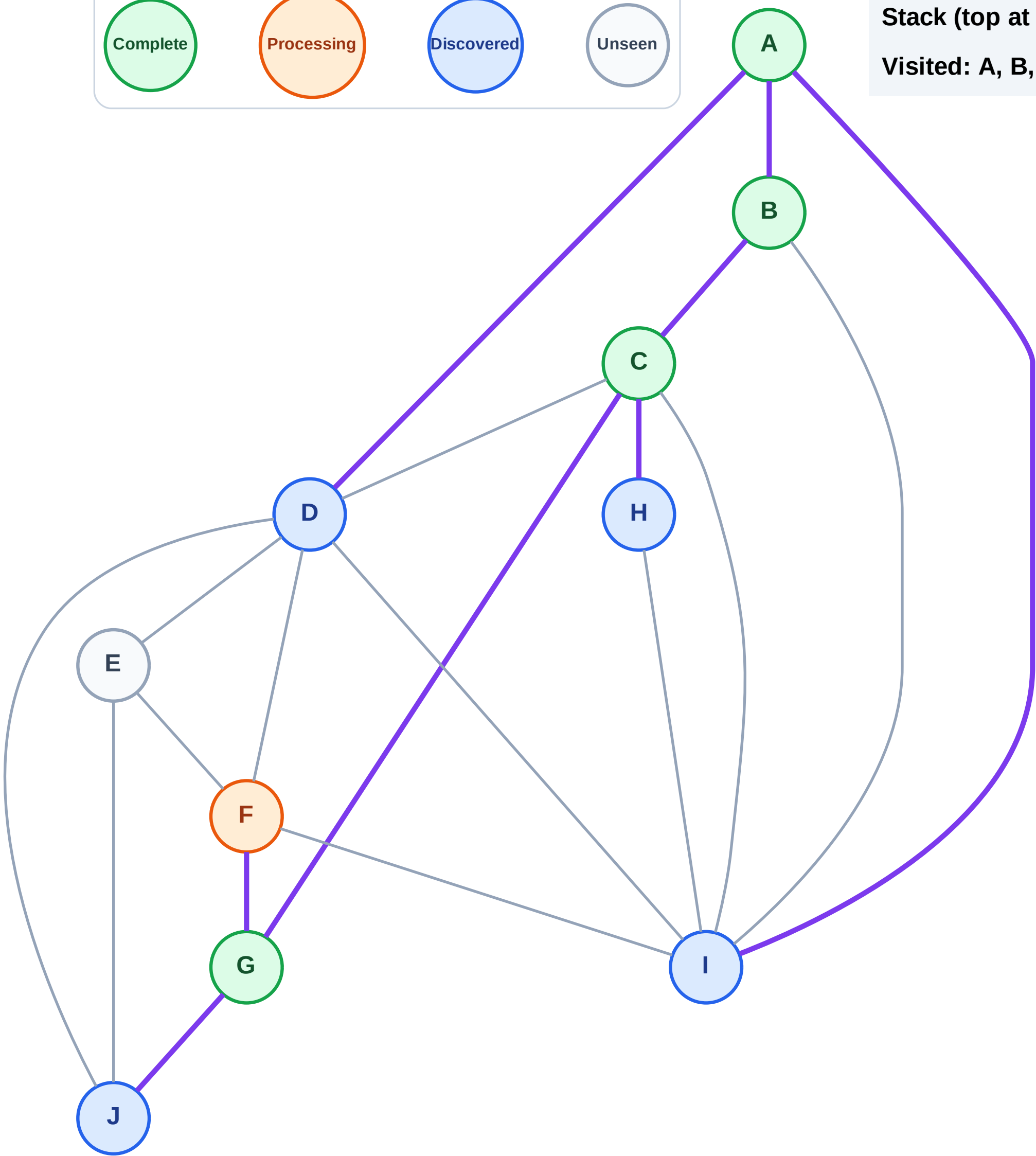
Step 10: Process node F

Legend



Stack (top at right): I | D | H | J

Visited: A, B, C, G



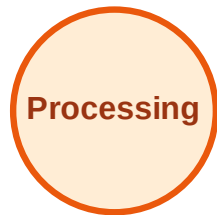
DFS Traversal

Step 11: Discover E from F

Legend



Complete



Processing



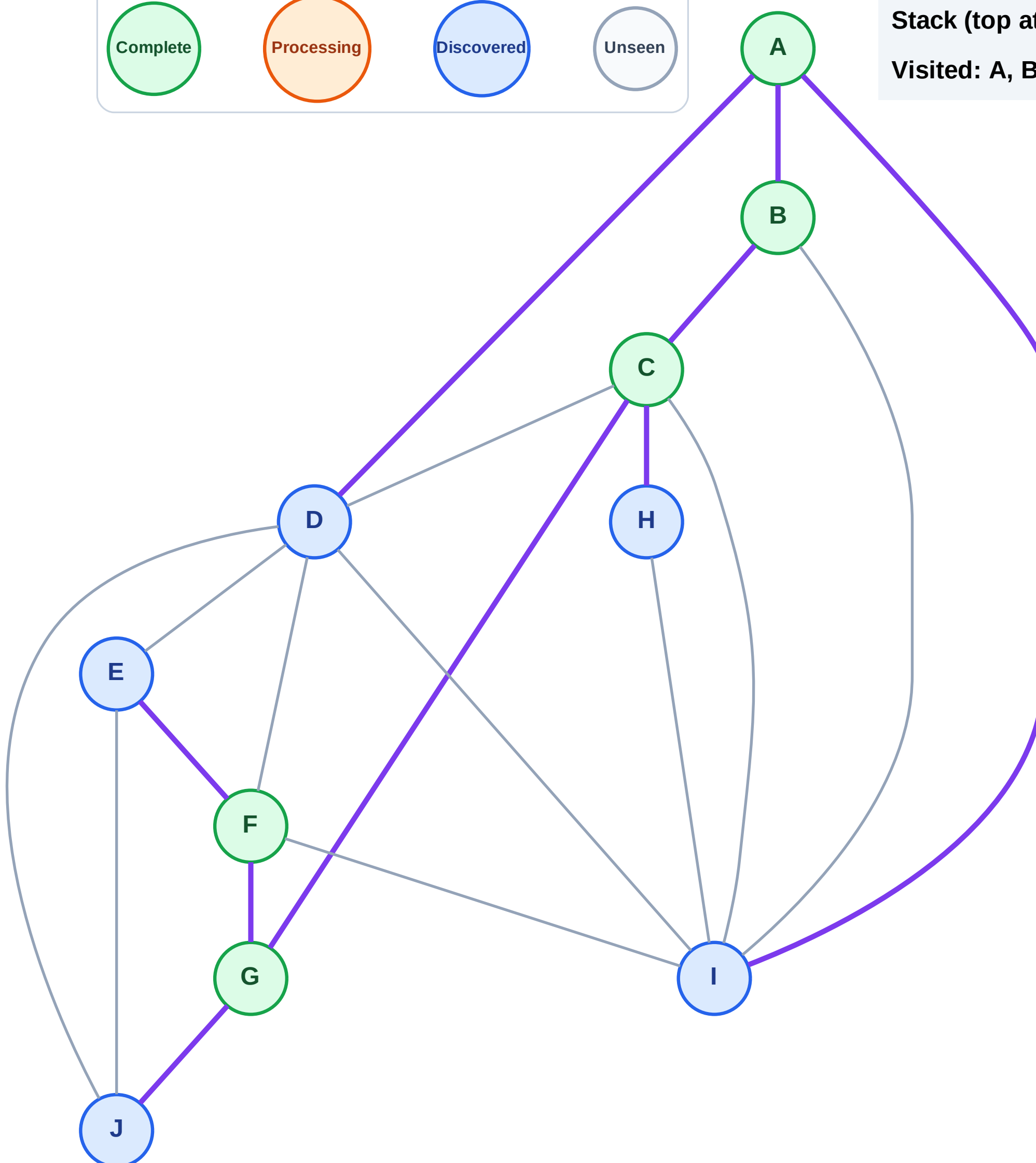
Discovered



Unseen

Stack (top at right): I | D | H | J | E

Visited: A, B, C, F, G



DFS Traversal

Step 12: Process node E

Legend

Complete

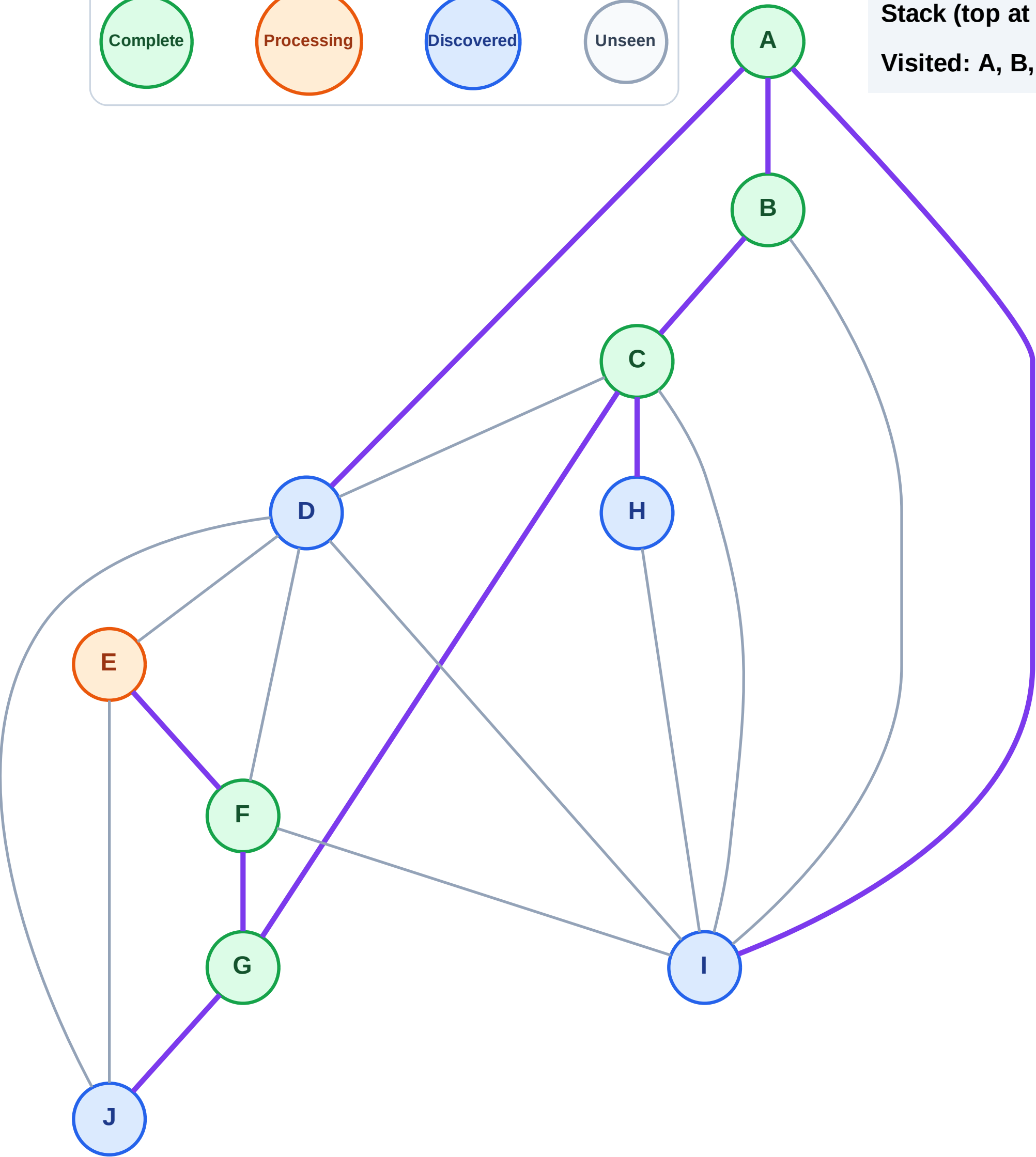
Processing

Discovered

Unseen

Stack (top at right): I | D | H | J

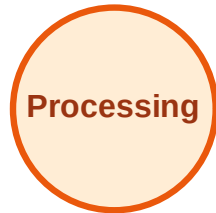
Visited: A, B, C, F, G



DFS Traversal

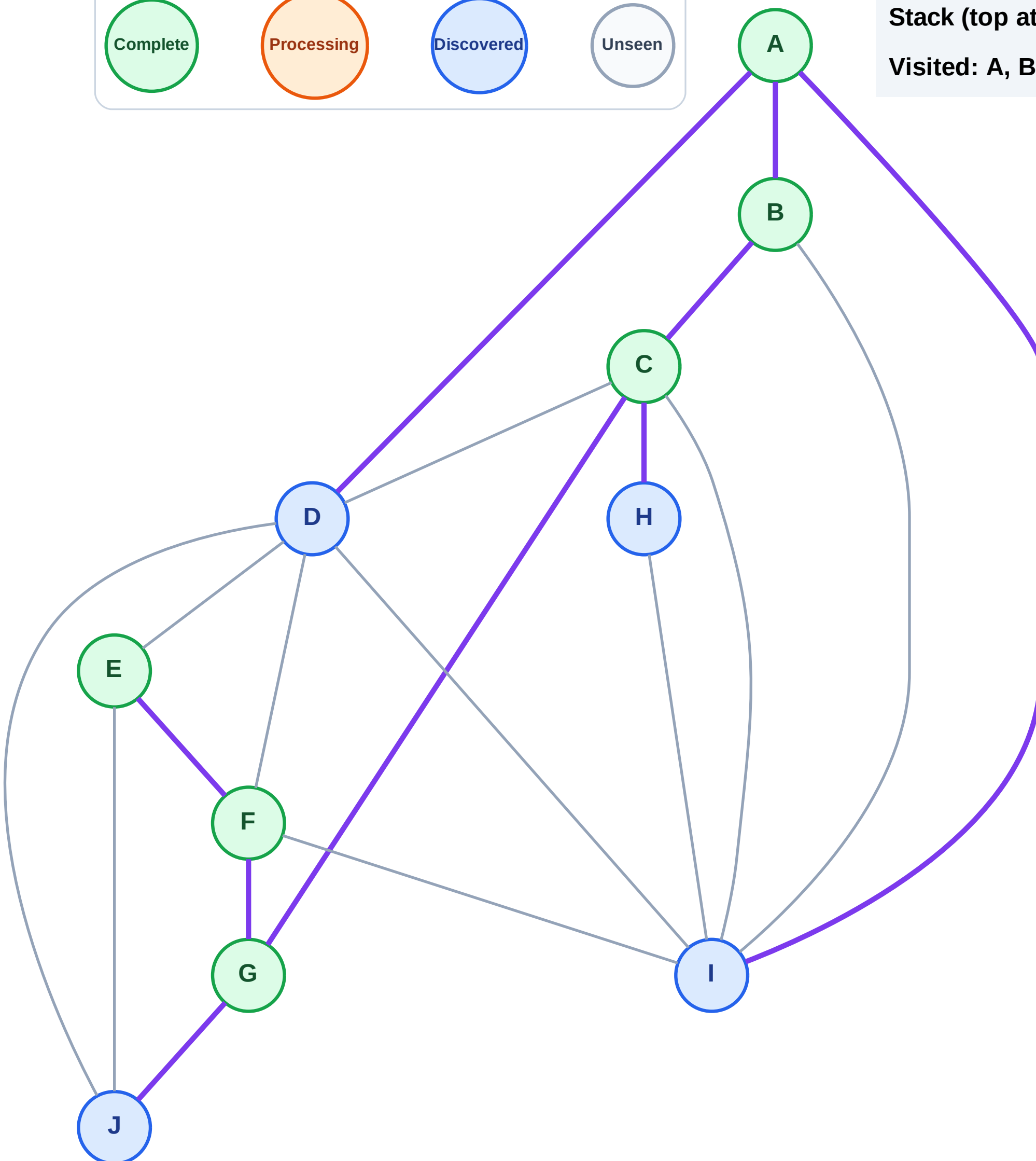
Step 13: No new neighbors from E

Legend



Stack (top at right): I | D | H | J

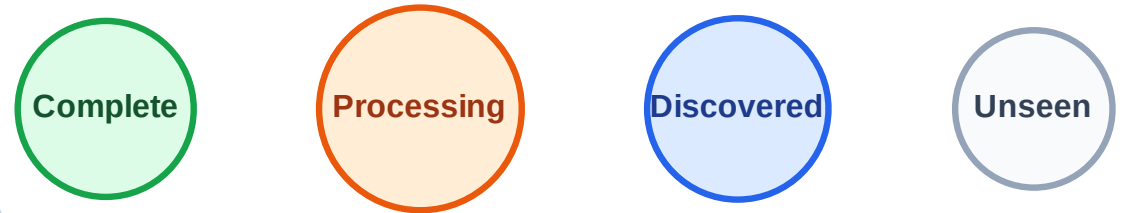
Visited: A, B, C, E, F, G



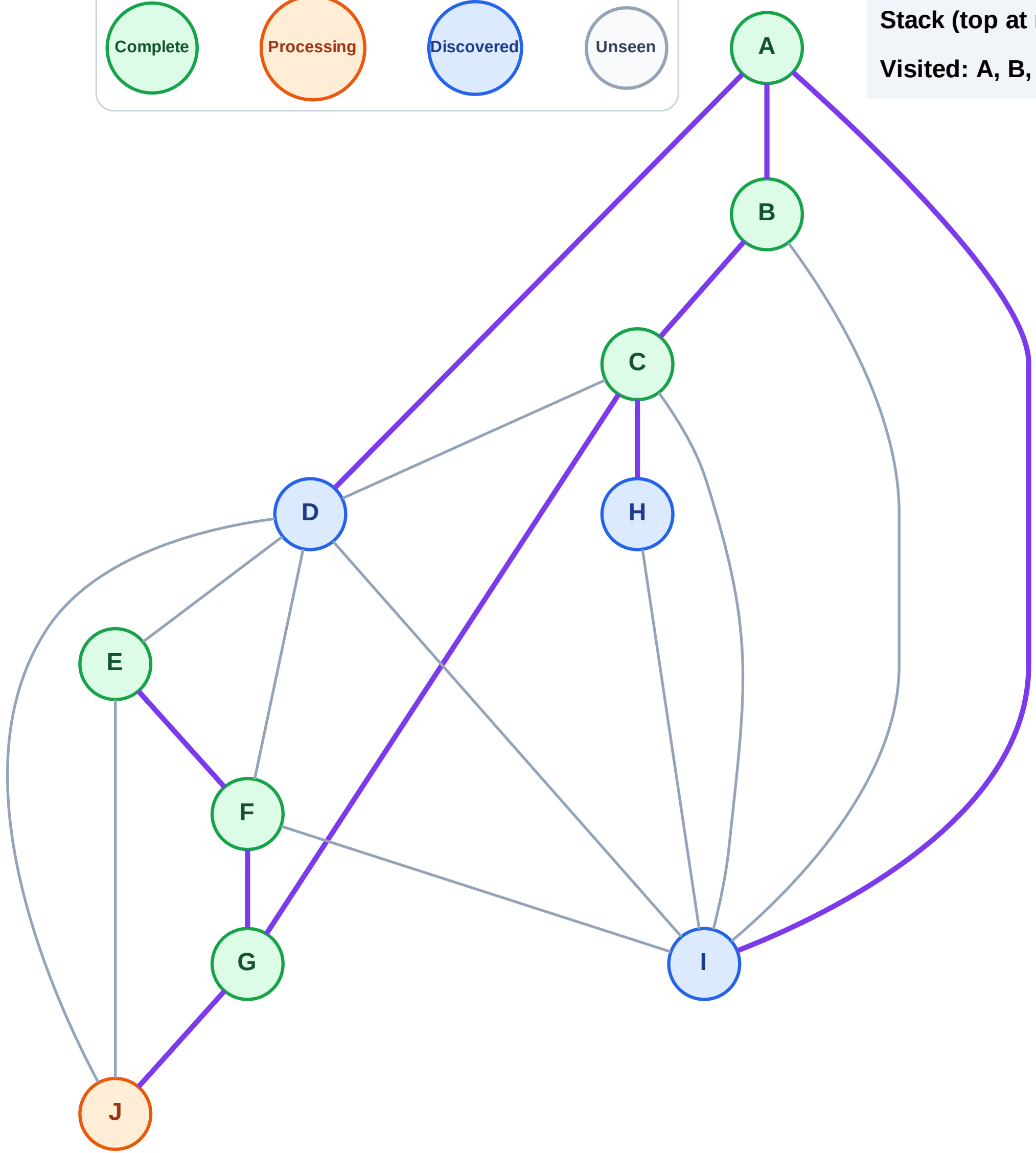
DFS Traversal

Step 14: Process node J

Legend

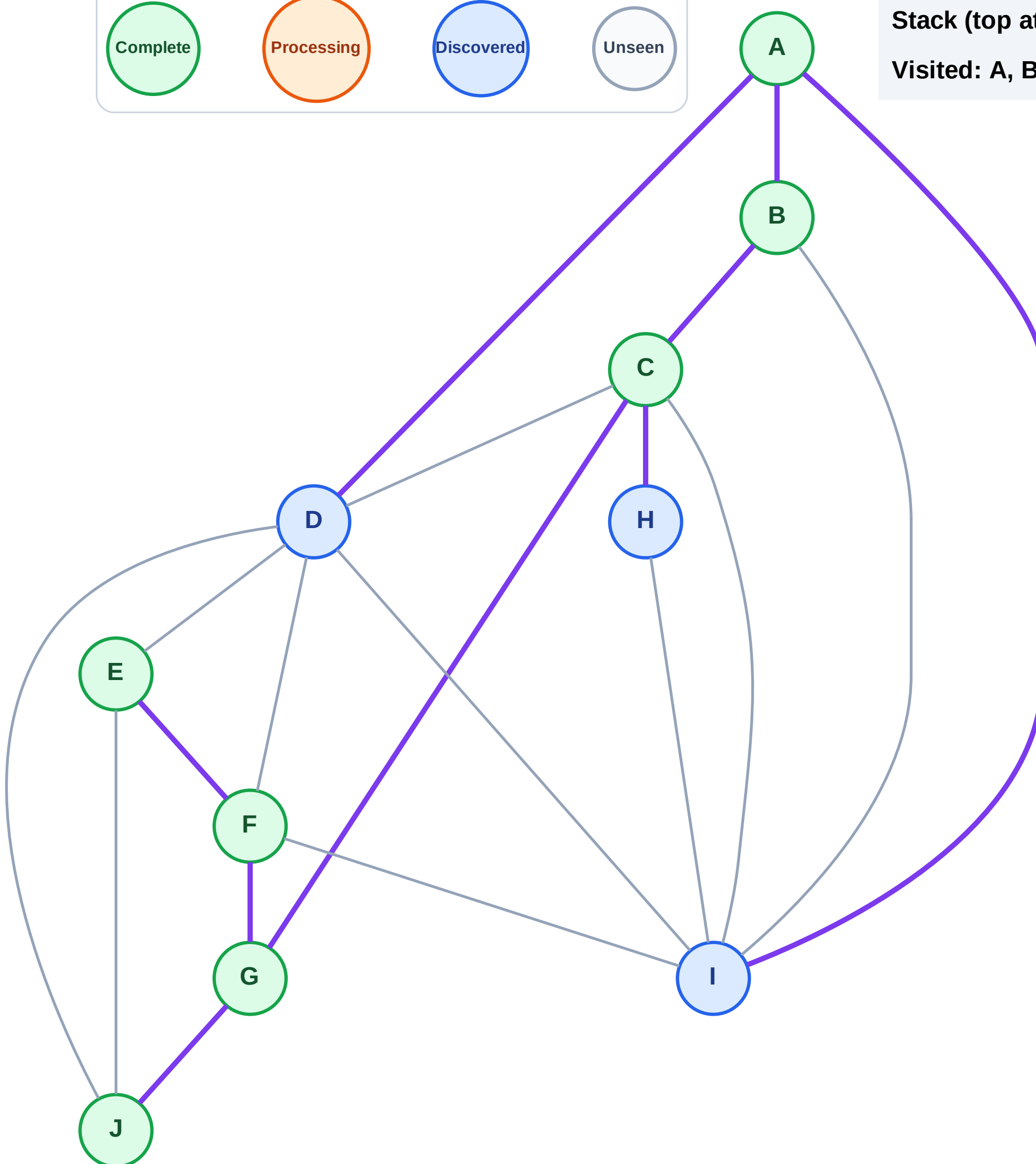
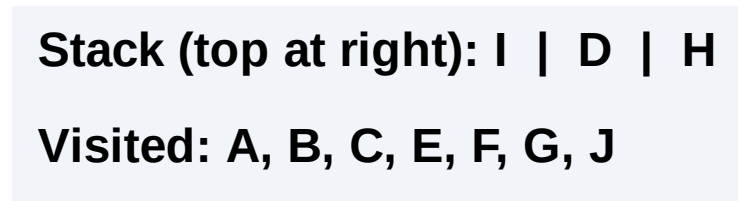


Stack (top at right): I | D | H
Visited: A, B, C, E, F, G



Step 15: No new neighbors from J

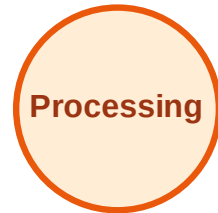
Step 15: No new neighbors from J



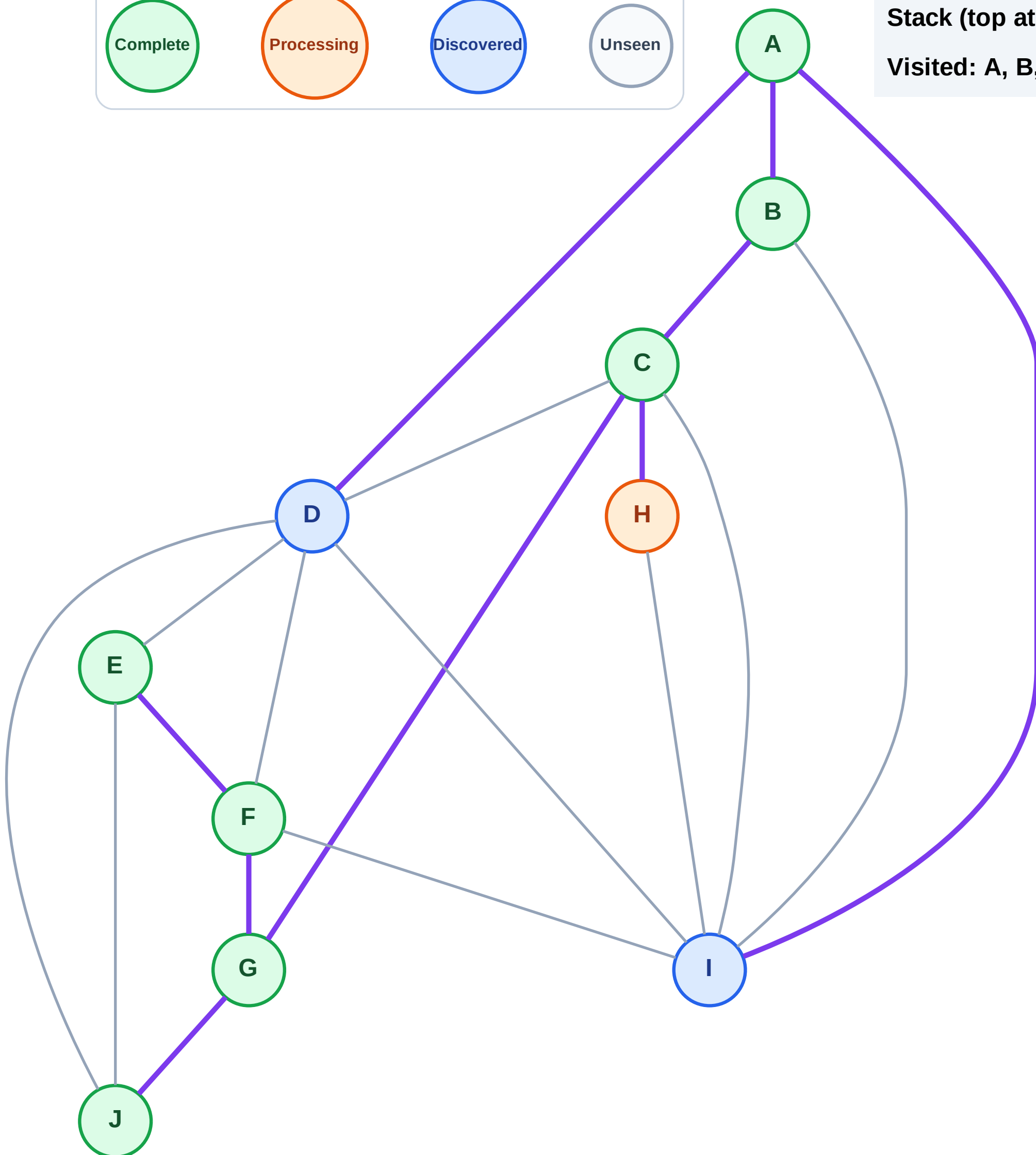
DFS Traversal

Step 16: Process node H

Legend



Stack (top at right): I | D
Visited: A, B, C, E, F, G, J



DFS Traversal

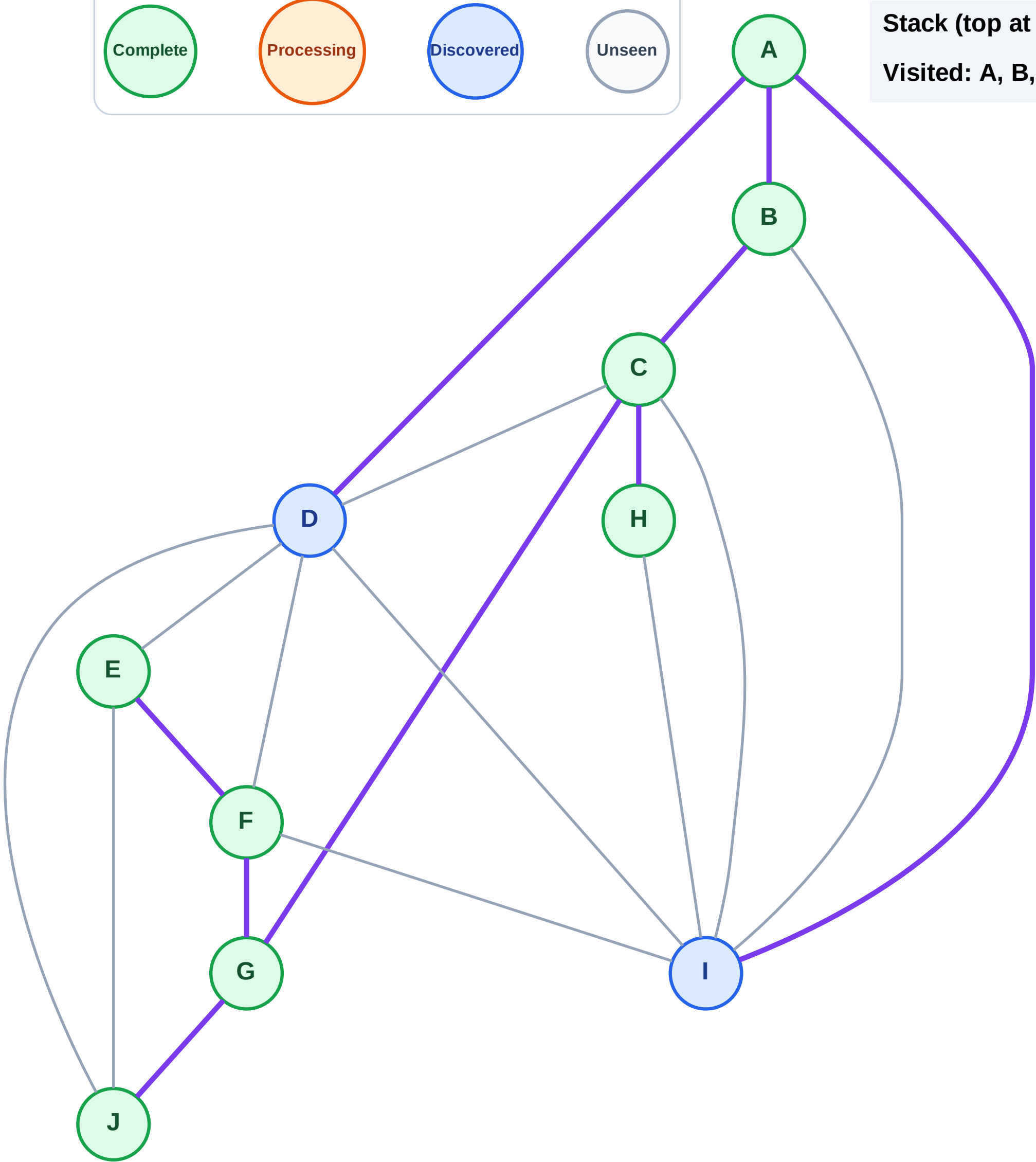
Step 17: No new neighbors from H

Legend



Stack (top at right): I | D

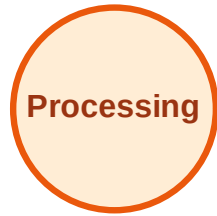
Visited: A, B, C, E, F, G, H, J



DFS Traversal

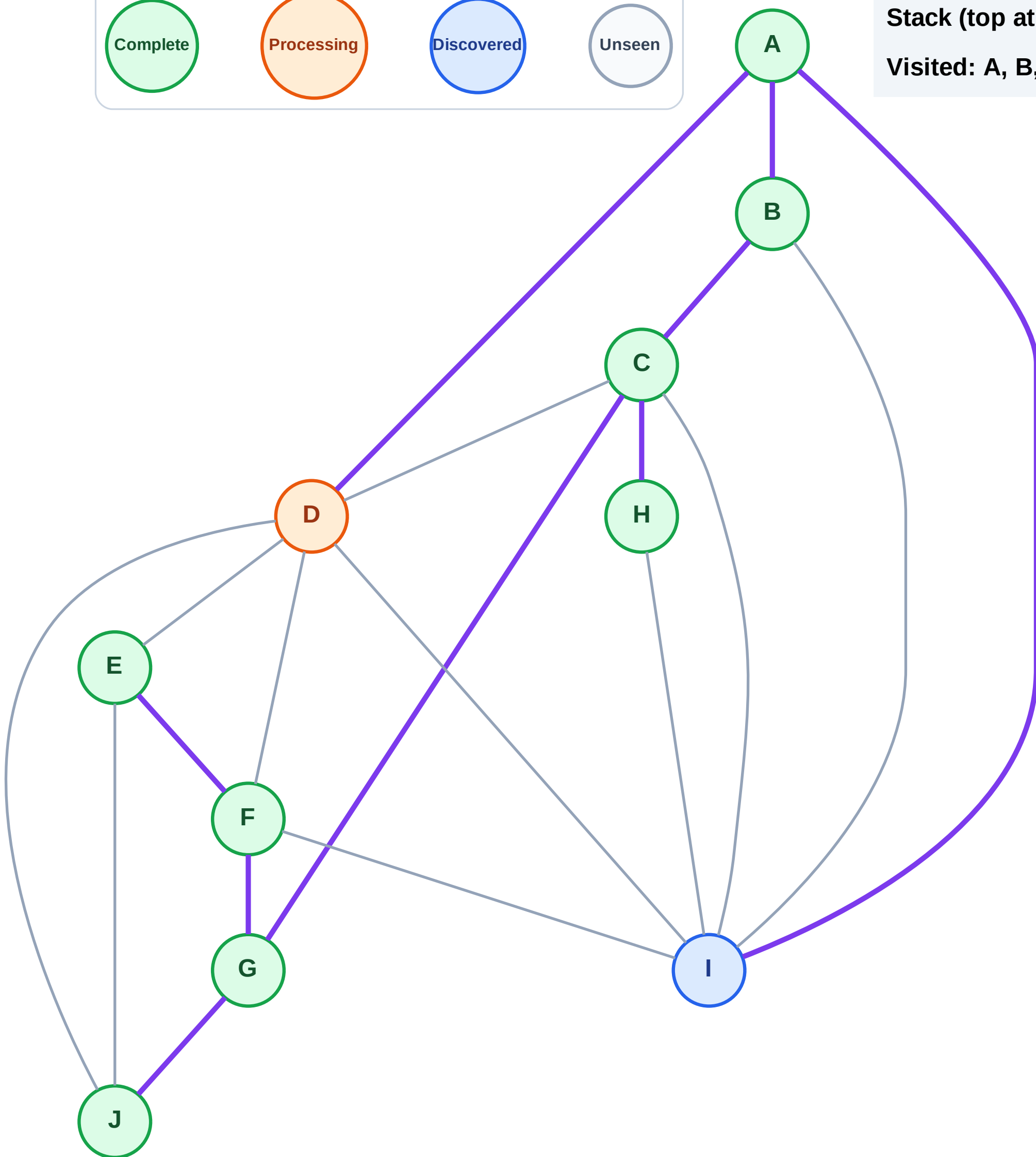
Step 18: Process node D

Legend



Stack (top at right): I

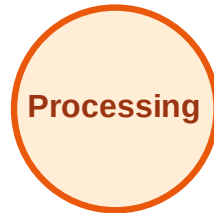
Visited: A, B, C, E, F, G, H, J



DFS Traversal

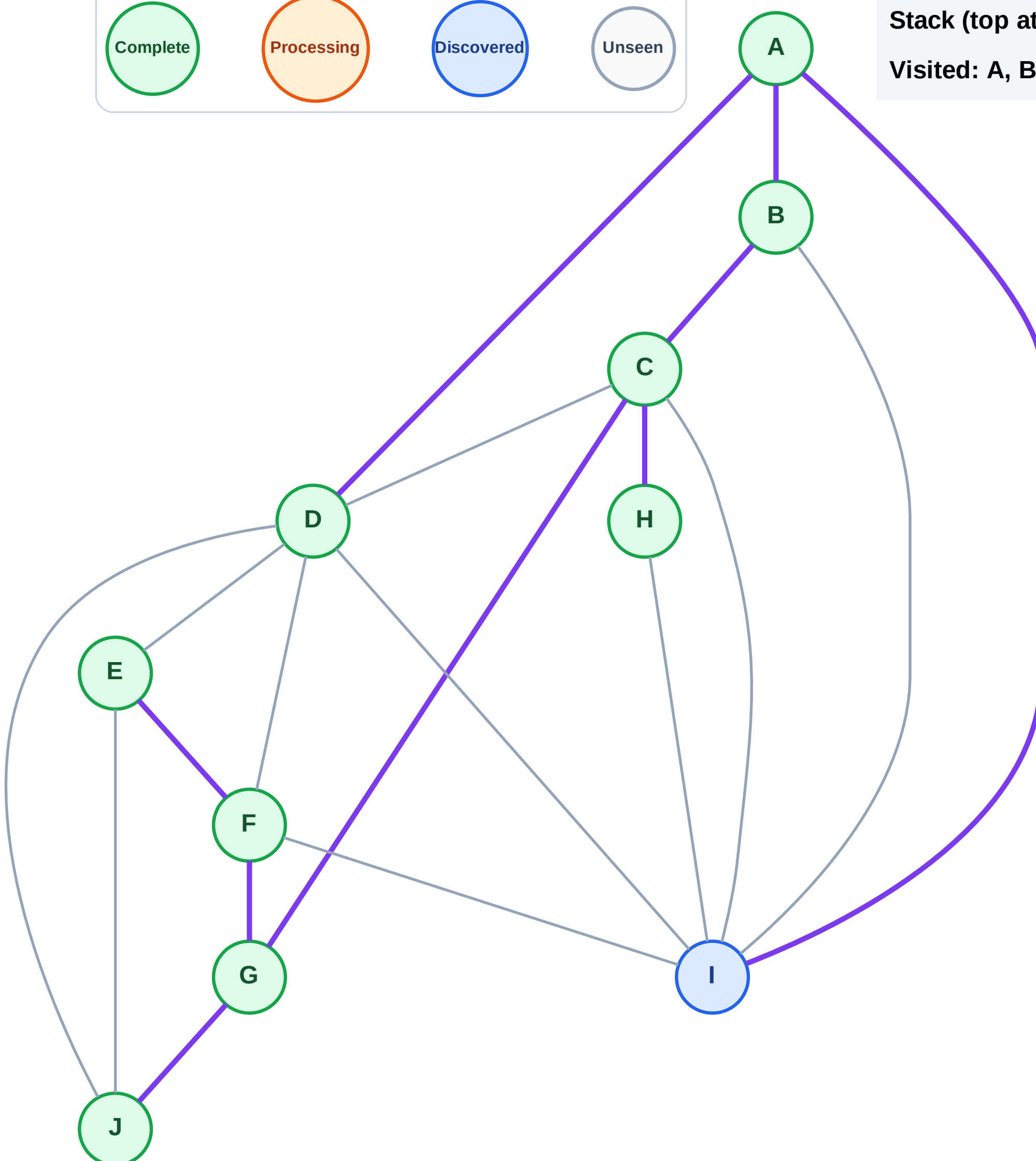
Step 19: No new neighbors from D

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 20: Process node I

Legend

Complete

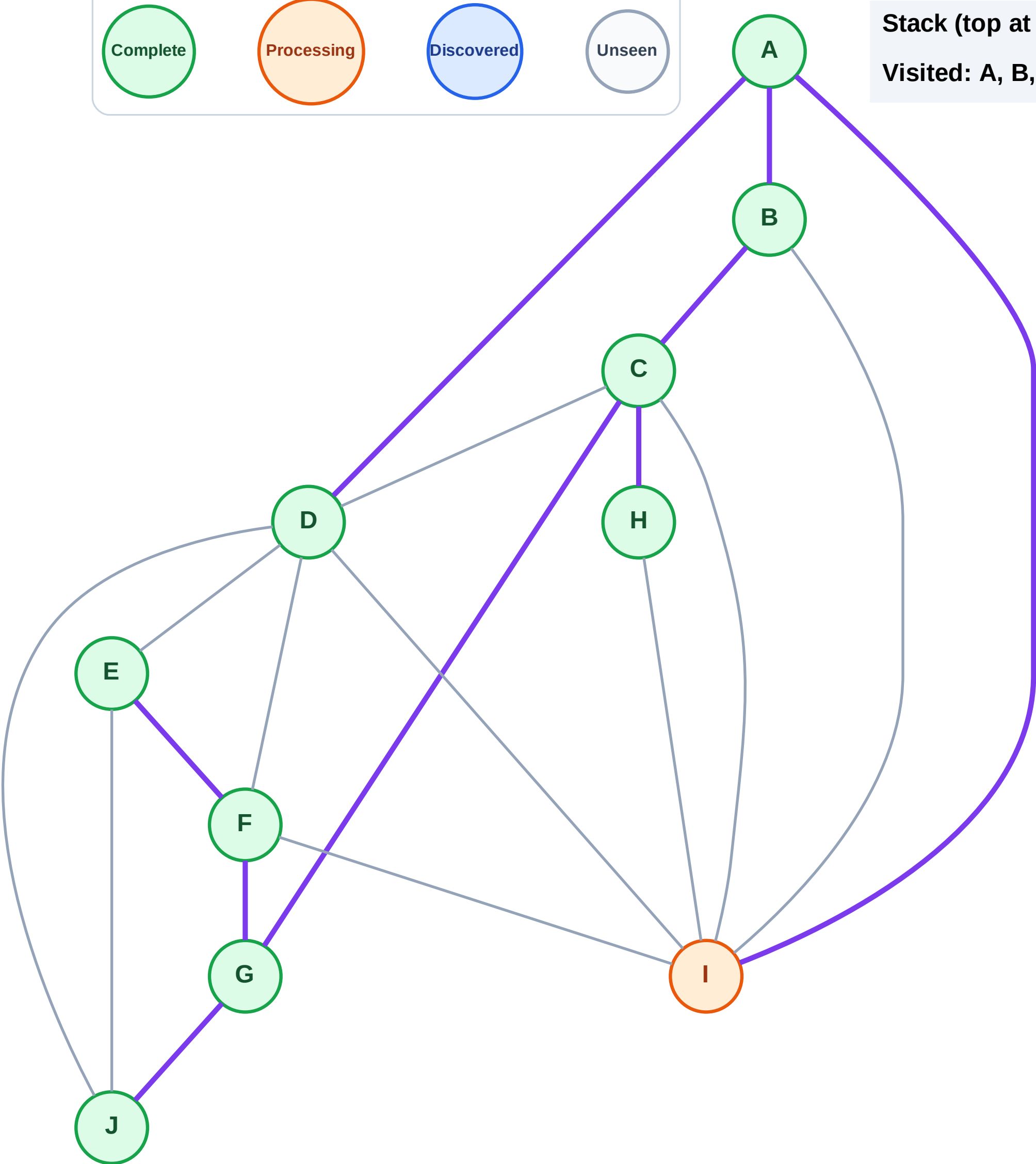
Processing

Discovered

Unseen

Stack (top at right): (empty)

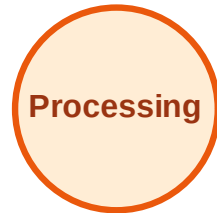
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

